Econometrics: Advanced Methods

Short lecture notes for Research Metrics 2 (MQE, LMU Munich), Winter 2026/27

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Preface

These notes accompany the course *Econometrics: Advanced Methods (Research Metrics 2)* in the Munich Graduate School of Economics MQE programme. They are deliberately short. Each chapter states the key definitions and results, works through one or two examples in detail, and ends with a computer tutorial in R. Proofs are sketched where the argument is instructive and otherwise referenced.

The notes complement the lecture slides; they do not replace the textbooks listed below.

What the course covers

The objective is to introduce fundamental estimation methods in econometrics. We spend a fair amount of time on the theoretical properties of the estimators and also discuss practical aspects of implementation. The course is not a course on causality or identification, and it is not an R course, although both appear throughout.

Chapter	Topic	Computer tutorial
1	Maximum likelihood	CT 1: logit by hand, PML vs. ML
2	Method of moments and GMM, including IV	CT 2: GMM estimation of an ATE
3	Simulation-based estimation	CT 3: Monte Carlo integration, Halton draws, quadrature
4	Nonparametric density and regression estimation	CT 4: kernel density, local linear regression
5	Introduction to statistical learning	CT 5: ridge, lasso, elastic net
A	Large-sample asymptotics (assumed known; for reference)	
B	A short R primer	CT 0

Literature

- Hansen, B. E. (2022). *Econometrics*. Princeton University Press (B. E. Hansen 2022a); and (2022). *Probability and Statistics for Economists*. Princeton University Press (B. E. Hansen 2022b).
- Cameron, A. C. and Trivedi, P. K. (2005). *Microeconometrics: Methods and Applications*. Cambridge University Press (Cameron and Trivedi 2005).
- Wooldridge, J. M. (2010). *Econometric Analysis of Cross Section and Panel Data*, 2nd ed. MIT Press (Wooldridge 2010).

- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2023). *An Introduction to Statistical Learning*, 2nd ed. Springer (James et al. 2023).
- Casella, G. and Berger, R. L. (2002). *Statistical Inference*, 2nd ed. Duxbury (Casella and Berger 2002).

Chapter-specific readings are given at the start of each chapter.

Software

All tutorials use R. Install a recent version of R and an editor (RStudio or Positron), then install the packages used in the notes:

```
install.packages(c("sandwich", "lmtest", "MASS", "nleqslv",
                   "randtoolbox", "statmod", "glmnet", "haven"))
```

Every tutorial can be run from top to bottom. Data sets are in the `data/` folder of the course material on Moodle.

How to use the tutorials

Each tutorial first states the problems. Try them yourself before opening the solution boxes. Solutions are released after the tutorial session.

Conventions

Vectors and matrices are in bold, $\boldsymbol{\theta}$ denotes a generic parameter vector and $\boldsymbol{\theta}_0$ its true value. ln is the natural logarithm. $\xrightarrow{p}$ and $\xrightarrow{d}$ denote convergence in probability and in distribution.

Part I

Estimation principles

1 Maximum Likelihood

Readings

Wooldridge (2010), Ch. 13.1–13.7 and 13.11; Cameron and Trivedi (2005), Ch. 5.6–5.7; B. E. Hansen (2022b), Ch. 10; B. E. Hansen (2022a), Ch. 22, 25 and 26.10. Applications: Smith (2009) and Winkelmann (2004).

Maximum likelihood (ML) is the workhorse estimator for fully specified parametric models. This chapter introduces the likelihood principle, derives the large-sample properties of the MLE, discusses what happens when the assumed density is wrong, and closes with the classical trinity of tests.

1.1 Parametric models

We first consider unconditional distributions. A **parametric model** for a random variable Y is a complete probability function $f(y \mid \boldsymbol{\theta})$ that depends on an unknown parameter vector $\boldsymbol{\theta}$ in a **parameter space** Θ. We use the same notation for probability mass functions (discrete Y) and densities (continuous Y). For example, $Y \sim N(\mu, \sigma^2)$ has density $f(y \mid \mu, \sigma^2) = \sigma^{-1}\phi((y-\mu)/\sigma)$ with $\phi(z) = (2\pi)^{-1/2}\exp(-z^2/2)$, $\mu \in \mathbb{R}$ and $\sigma^2 > 0$. A parametric model need not be a textbook distribution; it can be built for a specific application.

Definition 1.1 (Model)**.** A model for a random sample is the assumption that $Y_1, \ldots, Y_n$ are i.i.d. with a known density (or mass) function $f(y \mid \boldsymbol{\theta})$ and unknown parameter $\boldsymbol{\theta} \in \Theta$.

Definition 1.2 (Correct specification)**.** A model is **correctly specified** if there exists a unique $\boldsymbol{\theta}_0 \in \Theta$ such that $f(y \mid \boldsymbol{\theta}_0) = f(y)$, the true density. $\boldsymbol{\theta}_0$ is the **true parameter value**. Uniqueness means that no other $\boldsymbol{\theta} \in \Theta$ satisfies $f(y \mid \boldsymbol{\theta}) = f(y \mid \boldsymbol{\theta}_0)$. A model is **misspecified** if no $\boldsymbol{\theta} \in \Theta$ satisfies $f(y \mid \boldsymbol{\theta}) = f(y)$.

Example 1.1 (Correct and incorrect models)**.** Let the true density be $f(y) = 2e^{-2y}$, $y > 0$.

- The exponential model $f(y \mid \lambda) = \lambda^{-1}e^{-y/\lambda}$ is correctly specified with $\lambda_0 = 1/2$.
- The gamma model $f(y \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1}e^{-\beta y}$ is correctly specified with $(\alpha_0, \beta_0) = (1, 2)$.
- The log-normal model $f(y \mid \theta, \nu) = (2\pi\nu)^{-1/2}y^{-1}\exp\{-(\ln y - \theta)^2/(2\nu)\}$ is misspecified: no (θ, ν) reproduces $2e^{-2y}$.

Uniqueness matters. A two-component normal mixture $p\,\phi_{\sigma_1}(y-\mu_1) + (1-p)\,\phi_{\sigma_2}(y-\mu_2)$ contains the standard normal $\phi(y)$, but at many parameter values: $(p, 0, 1, 0, 1)$ for any p, $(1, 0, 1, \mu_2, \sigma_2^2)$ for any (μ_2, σ_2^2), and so on. The model contains the truth, but $\boldsymbol{\theta}_0$ is not identified, so it is not correctly specified in the sense of Definition 1.2.

1.2 The likelihood function and the MLE

The **likelihood** is the joint density of the sample, evaluated at the observed data and viewed as a function of $\boldsymbol{\theta}$. Under independence it factorises, and under identical distributions all factors share the same form:

$$L_n(\boldsymbol{\theta}) = f(y_1, \dots, y_n \mid \boldsymbol{\theta}) = \prod_{i=1}^{n} f(y_i \mid \boldsymbol{\theta}), \qquad \ell_n(\boldsymbol{\theta}) \equiv \ln L_n(\boldsymbol{\theta}) = \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{\theta}).$$

Definition 1.3 (Maximum likelihood estimator)**.** The MLE is the parameter value that maximises the likelihood,

$$\hat{\boldsymbol{\theta}}_n = \arg\max_{\boldsymbol{\theta} \in \Theta} L_n(\boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta} \in \Theta} \frac{1}{n} \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{\theta}).$$

The second equality holds because $\ln(\cdot)$ is strictly increasing and scaling by $1/n$ does not change the maximiser.

The MLE is the parameter value under which the observed sample is most likely. Working with the average log-likelihood is convenient: it is a sample mean, so laws of large numbers and central limit theorems apply directly.

Example 1.2 (Bernoulli)**.** Let $Y_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\pi)$, so $f(y \mid \pi) = \pi^y (1-\pi)^{1-y}$ for $y \in \{0, 1\}$. Then

$$\ell_n(\pi) = \sum_{i=1}^{n} \left[y_i \ln \pi + (1 - y_i) \ln(1 - \pi) \right] = n \left[\bar{y} \ln \pi + (1 - \bar{y}) \ln(1 - \pi) \right].$$

The first-order condition $n\bar{y}/\pi - n(1-\bar{y})/(1-\pi) = 0$ gives $\hat{\pi} = \bar{y}$. Figure 1.1 plots the likelihood for two samples of size five.

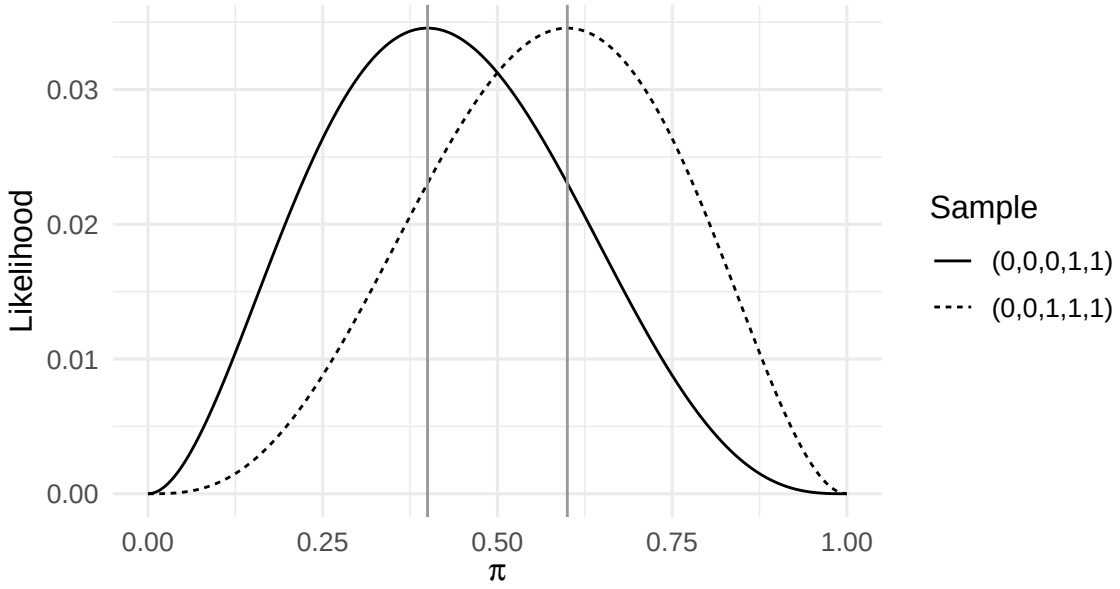


Figure 1.1: Bernoulli likelihood for the samples (0,0,0,1,1) and (0,0,1,1,1). The maximum is at the sample mean.

In practice we find the MLE in one of three ways:

1. **Solve the first-order condition** $\partial \ell_n(\boldsymbol{\theta})/\partial \boldsymbol{\theta} = \mathbf{0}$ and check the second-order condition. The FOC is necessary but not sufficient, and it only locates interior extrema; a maximum on the boundary of Θ need not satisfy it.
2. **Direct argument**, for instance by finding an upper bound of the likelihood that is attained at a unique point. The uniform distribution on $[0, \theta]$ is the standard case: the likelihood $\theta^{-n}\mathbf{1}\{\theta \geq \max_i x_i\}$ is maximised at the largest observation. Problem Set 1 treats a shifted exponential distribution, where the same argument applies.
3. **Numerical optimisation** when no closed form exists (logit, probit, Tobit, ...). We then have to worry about convergence to a local rather than a global maximum.

Example 1.3 (Exponential)**.** Let $f(y \mid \lambda) = \lambda^{-1} e^{-y/\lambda}$, $y > 0$. The log-likelihood and its derivatives are

$$\ell_n(\lambda) = -n \ln \lambda - \frac{n\bar{y}}{\lambda}, \qquad \frac{d\ell_n}{d\lambda} = -\frac{n}{\lambda} + \frac{n\bar{y}}{\lambda^2}, \qquad \frac{d^2\ell_n}{d\lambda^2} = \frac{n}{\lambda^2} - \frac{2n\bar{y}}{\lambda^3}.$$

The unique root of the FOC is $\hat{\lambda} = \bar{y}$, and $d^2\ell_n(\hat{\lambda})/d\lambda^2 = -n/\bar{y}^2 < 0$ confirms a maximum.

Theorem 1.1 (Invariance)**.** *If $\hat{\boldsymbol{\theta}}$ is the MLE of $\boldsymbol{\theta} \in \mathbb{R}^m$, then for any function $\boldsymbol{\beta} = h(\boldsymbol{\theta}) \in \mathbb{R}^l$ the MLE of $\boldsymbol{\beta}$ is $\hat{\boldsymbol{\beta}} = h(\hat{\boldsymbol{\theta}})$.*

In Example 1.3, reparametrise with the rate $\beta = 1/\lambda$. Then $\ell_n(\beta) = n \ln \beta - \beta n \bar{y}$, maximised at $\hat{\beta} = 1/\bar{y} = h(\hat{\lambda})$, as the theorem predicts. Most estimators do not share this property; an unbiased estimator of λ, for example, does not deliver an unbiased estimator of $1/\lambda$.

1.3 Score, Hessian and information

Assume $f(y \mid \boldsymbol{\theta})$ is twice differentiable in $\boldsymbol{\theta} \in \mathbb{R}^k$.

Definition 1.4 (Score and Hessian)**.** The **score** is the $k \times 1$ gradient of the log-likelihood and the **Hessian** is the $k \times k$ matrix of second derivatives:

$$S_n(\boldsymbol{\theta}) = \frac{\partial \ell_n(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \sum_{i=1}^{n} \underbrace{\frac{\partial \ln f(y_i \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}}_{S_i(\boldsymbol{\theta})}, \qquad H_n(\boldsymbol{\theta}) = \frac{\partial^2 \ell_n(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'} = \sum_{i=1}^{n} \underbrace{\frac{\partial^2 \ln f(y_i \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'}}_{H_i(\boldsymbol{\theta})}.$$

The score measures how sensitive the log-likelihood is to the parameters; at an interior MLE, $S_n(\hat{\boldsymbol{\theta}}) = \mathbf{0}$. The Hessian is symmetric and measures curvature. If $H_n(\boldsymbol{\theta})$ is negative definite for all $\boldsymbol{\theta}$, ℓ_n is strictly concave and a maximiser, if it exists, is unique. (The converse does not hold: a strictly concave function can have a singular Hessian at some points.) Both are sums over observations because the log-likelihood is.

Theorem 1.2 (Properties of the likelihood)**.** *Suppose the model is correctly specified, the support of Y does not depend on $\boldsymbol{\theta}$, and $\boldsymbol{\theta}_0$ is in the interior of Θ. Then*

(i) $\mathbb{E}[S_i(\boldsymbol{\theta}_0)] = \mathbf{0}$*;*

(ii) ***information matrix equality:*** $\mathbb{E}[S_i(\boldsymbol{\theta}_0) S_i(\boldsymbol{\theta}_0)'] = -\mathbb{E}[H_i(\boldsymbol{\theta}_0)] \equiv \mathcal{I}_1(\boldsymbol{\theta}_0)$.

$\mathcal{I}_1$ is the (Fisher) information of a single observation and $\mathcal{I}(\boldsymbol{\theta}_0) = n\mathcal{I}_1(\boldsymbol{\theta}_0)$ that of the sample. Because the score has mean zero, the left side of (ii) is the variance of the score. Part (i) follows from differentiating $\int f(y \mid \boldsymbol{\theta})\, dy = 1$ under the integral sign; part (ii) from differentiating once more. The support condition guarantees that this exchange of differentiation and integration is allowed. The uniform $U[0, \theta]$ violates it, which is why its MLE behaves non-standardly.

Proof of Theorem 1.2

Write $f = f(y \mid \boldsymbol{\theta})$ and use $\frac{\partial f}{\partial \boldsymbol{\theta}} = \frac{\partial \ln f}{\partial \boldsymbol{\theta}} f$.
(i) A density integrates to one for every $\boldsymbol{\theta}$: $\int f(y \mid \boldsymbol{\theta})\, dy = 1$. Differentiate with respect to $\boldsymbol{\theta}$ and exchange derivative and integral:

$$\mathbf{0} = \int \frac{\partial f}{\partial \boldsymbol{\theta}}\, dy = \int \frac{\partial \ln f}{\partial \boldsymbol{\theta}}\, f\, dy = \mathbb{E}\Big[\frac{\partial \ln f(y \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\Big]. \qquad (*)$$

At $\boldsymbol{\theta} = \boldsymbol{\theta}_0$ the expectation is over the true distribution, so $\mathbb{E}[S_i(\boldsymbol{\theta}_0)] = \mathbf{0}$.
(ii) $(*)$ holds for every $\boldsymbol{\theta}$, so we can differentiate it with respect to $\boldsymbol{\theta}'$. By the product rule,

$$\mathbf{0} = \int \frac{\partial^2 \ln f}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'}\, f\, dy + \int \frac{\partial \ln f}{\partial \boldsymbol{\theta}} \frac{\partial f}{\partial \boldsymbol{\theta}'}\, dy = \mathbb{E}\Big[\frac{\partial^2 \ln f}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'}\Big] + \mathbb{E}\Big[\frac{\partial \ln f}{\partial \boldsymbol{\theta}} \frac{\partial \ln f}{\partial \boldsymbol{\theta}'}\Big],$$

using $\frac{\partial f}{\partial \boldsymbol{\theta}'} = \frac{\partial \ln f}{\partial \boldsymbol{\theta}'} f$ in the second term. Rearranging gives $\mathbb{E}[S_i S_i'] = -\mathbb{E}[H_i]$. □

Theorem 1.3 (Cramér–Rao lower bound)**.** *Under the same regularity conditions, any unbiased estimator $\tilde{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}_0$ satisfies* $\operatorname{Var}(\tilde{\boldsymbol{\theta}}) \succeq \mathcal{I}(\boldsymbol{\theta}_0)^{-1}$.

Example 1.4 (Bernoulli, continued)**.** For one observation, $S_i(\pi) = \dfrac{y_i - \pi}{\pi(1-\pi)}$ and $H_i(\pi) = -\dfrac{y_i}{\pi^2} - \dfrac{1 - y_i}{(1-\pi)^2}$. At π_0: $\mathbb{E}[S_i] = 0$, $\operatorname{Var}(S_i) = \dfrac{\pi_0(1-\pi_0)}{\pi_0^2(1-\pi_0)^2} = \dfrac{1}{\pi_0(1-\pi_0)}$, and $-\mathbb{E}[H_i] = \dfrac{1}{\pi_0} + \dfrac{1}{1-\pi_0} = \dfrac{1}{\pi_0(1-\pi_0)}$. The information equality holds and the CRLB is $\pi_0(1-\pi_0)/n$, which $\hat{\pi} = \bar{y}$ attains.

1.3.1 Conditional likelihood

Econometric models are usually about the distribution of y given covariates $\boldsymbol{x}$. Write the joint density as $f(y, \boldsymbol{x} \mid \boldsymbol{\delta}) = f(y \mid \boldsymbol{x}; \boldsymbol{\theta})\, f(\boldsymbol{x} \mid \boldsymbol{\gamma})$ with $\boldsymbol{\delta} = (\boldsymbol{\theta}', \boldsymbol{\gamma}')'$. If $\boldsymbol{\theta}$ and $\boldsymbol{\gamma}$ share no elements and are not linked by restrictions,

$$\ell_n(\boldsymbol{\delta}) = \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\theta}) + \sum_{i=1}^{n} \ln f(\boldsymbol{x}_i \mid \boldsymbol{\gamma}),$$

and $\boldsymbol{\theta}$ can be estimated from the **conditional log-likelihood** $\sum_i \ln f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\theta})$ alone, leaving the distribution of $\boldsymbol{x}$ unspecified. All results above carry over with expectations taken over the joint distribution of $(y, \boldsymbol{x})$.

1.4 Asymptotic properties of the MLE

Theorem 1.4 (Consistency (Wooldridge 2010, Thm. 13.1))**.** *Suppose (a) the conditional density $f(y \mid \boldsymbol{x}; \boldsymbol{\theta})$ is correctly specified; (b) $\boldsymbol{\theta}_0$ is globally identified: for every $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0$ in Θ, $P\{f(y \mid \boldsymbol{x}; \boldsymbol{\theta}) \neq f(y \mid \boldsymbol{x}; \boldsymbol{\theta}_0)\} > 0$; (c) Θ is compact and the log-likelihood is continuous in $\boldsymbol{\theta}$ with a dominance condition. Then $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$.*

The idea: $\frac{1}{n}\ell_n(\boldsymbol{\theta}) \xrightarrow{p} \mathbb{E}[\ln f(y \mid \boldsymbol{x}; \boldsymbol{\theta})]$ uniformly in $\boldsymbol{\theta}$, and the limit is uniquely maximised at $\boldsymbol{\theta}_0$ (this follows from the KLIC inequality in Section 1.6). The maximiser of the sample objective then converges to the maximiser of the limit.

Theorem 1.5 (Asymptotic normality and efficiency)**.** *Under additional regularity conditions (interior $\boldsymbol{\theta}_0$, twice continuously differentiable log-density, nonsingular $\mathcal{I}_1(\boldsymbol{\theta}_0)$),*

$$\sqrt{n}\,(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0},\, \mathcal{I}_1(\boldsymbol{\theta}_0)^{-1}).$$

The MLE is asymptotically efficient: its asymptotic variance attains the Cramér–Rao bound, and no consistent and asymptotically normal estimator has a smaller asymptotic variance under the same assumptions.

Sketch. A mean-value expansion of the FOC around $\boldsymbol{\theta}_0$ gives $\mathbf{0} = S_n(\hat{\boldsymbol{\theta}}) = S_n(\boldsymbol{\theta}_0) + H_n(\bar{\boldsymbol{\theta}})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$, so

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = \Big[-\tfrac{1}{n} H_n(\bar{\boldsymbol{\theta}}) \Big]^{-1} \tfrac{1}{\sqrt{n}} S_n(\boldsymbol{\theta}_0).$$

Here $\bar{\boldsymbol{\theta}}$ lies between $\hat{\boldsymbol{\theta}}$ and $\boldsymbol{\theta}_0$, $\bar{\boldsymbol{\theta}} = \boldsymbol{\theta}_0 + \lambda(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$ with $0 \le \lambda \le 1$ (row by row), so $\bar{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}_0$ because $\hat{\boldsymbol{\theta}}$ is consistent. By the LLN and consistency, $-\frac{1}{n} H_n(\bar{\boldsymbol{\theta}}) \xrightarrow{p} \mathcal{I}_1(\boldsymbol{\theta}_0) \equiv \boldsymbol{A}_0$. By the CLT and Theorem 1.2, $\frac{1}{\sqrt{n}} S_n(\boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{B}_0)$ with $\boldsymbol{B}_0 = \mathbb{E}[S_i S_i']$. For a product of a matrix $\boldsymbol{H}_n \xrightarrow{p} \boldsymbol{H}$ and a vector $\boldsymbol{b}_n \xrightarrow{d} N(\boldsymbol{\mu}, \boldsymbol{B})$, Slutsky's theorem gives $\boldsymbol{H}_n \boldsymbol{b}_n \xrightarrow{d} N(\boldsymbol{H}\boldsymbol{\mu}, \boldsymbol{H}\boldsymbol{B}\boldsymbol{H}')$ (Appendix A). With $\boldsymbol{H}_n = [-\frac{1}{n} H_n(\bar{\boldsymbol{\theta}})]^{-1}$ this yields the general form $N(\mathbf{0}, \boldsymbol{A}_0^{-1} \boldsymbol{B}_0 \boldsymbol{A}_0^{-1})$, which collapses to $\mathcal{I}_1^{-1}$ because $\boldsymbol{A}_0 = \boldsymbol{B}_0$ under correct specification. Keep this sandwich form in mind; it reappears in Section 1.6 and in GMM.

1.5 Estimating the variance matrix

In finite samples we use $\operatorname{Var}(\hat{\boldsymbol{\theta}}) \approx \frac{1}{n} \mathcal{I}_1(\boldsymbol{\theta}_0)^{-1} = \mathcal{I}(\boldsymbol{\theta}_0)^{-1}$ and replace $\mathcal{I}_1$ by a consistent estimator. Three options are available, plus a fourth that does not rely on the information equality.

Table 1.1: Variance estimators after ML

	Estimator of $\operatorname{Var}(\hat{\boldsymbol{\theta}})$	Requires
(i) Expected Hessian	$\Big[-\sum_i \mathbb{E}\big(H_i(\boldsymbol{\theta}) \mid \boldsymbol{x}_i\big) \Big]^{-1}_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}$	closed form of $\mathbb{E}(H_i \mid \boldsymbol{x}_i)$
(ii) Observed Hessian	$\Big[-H_n(\hat{\boldsymbol{\theta}}) \Big]^{-1}$	second derivatives (analytic or numerical)
(iii) Outer product of gradients (OPG, BHHH)	$\Big[\sum_i S_i(\hat{\boldsymbol{\theta}}) S_i(\hat{\boldsymbol{\theta}})' \Big]^{-1}$	first derivatives only
(iv) Robust (sandwich)	$[-H_n(\hat{\boldsymbol{\theta}})]^{-1} \Big[\sum_i S_i S_i' \Big] [-H_n(\hat{\boldsymbol{\theta}})]^{-1}$	both

(i)–(iii) are all consistent under correct specification and typically differ in finite samples. The observed Hessian (ii) is the default in most software. The OPG (iii) is the cheapest but has the poorest finite-sample behaviour. The sandwich (iv) remains valid when the density is misspecified, as we see next.

For the Bernoulli model, all three of (i)–(iii) coincide at $\hat{\pi} = \bar{y}$ and equal $\bar{y}(1-\bar{y})/n$. Verifying this is a short exercise: use $\sum_i (y_i - \bar{y})^2 = n\bar{y}(1-\bar{y})$ for binary y_i.

1.5.1 Example: the normal linear model

Let $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \mid \boldsymbol{x}_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Then

$$\ell_n(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^2.$$

Setting the derivatives to zero yields

$$\hat{\boldsymbol{\beta}}_{ML} = \Big(\sum_i \boldsymbol{x}_i\boldsymbol{x}_i'\Big)^{-1}\sum_i \boldsymbol{x}_i y_i, \qquad \hat{\sigma}^2_{ML} = \frac{1}{n}\sum_i (y_i - \boldsymbol{x}_i'\hat{\boldsymbol{\beta}})^2.$$

$\hat{\boldsymbol{\beta}}_{ML}$ is OLS. $\hat{\sigma}^2_{ML}$ lacks the degrees-of-freedom correction, so it is biased downward in finite samples, but the bias vanishes as $n \to \infty$.

1.6 Misspecification and pseudo-ML

What does the MLE estimate when the model is wrong? The answer uses a measure of distance between densities.

Definition 1.5 (Kullback–Leibler divergence)**.** For densities f and g, $\mathrm{KLIC}(f, g) = \int f(y)\ln\dfrac{f(y)}{g(y)}\,dy$.

KLIC is not symmetric, $\mathrm{KLIC}(f, g) \neq \mathrm{KLIC}(g, f)$ in general, so it is not a distance in the metric sense. Its key properties are $\mathrm{KLIC}(f, f) = 0$, $\mathrm{KLIC}(f, g) \geq 0$ and therefore $f = \arg\min_g \mathrm{KLIC}(f, g)$. The second follows from Jensen's inequality, since ln is concave:

$$-\mathrm{KLIC}(f, g) = \mathbb{E}_f\Big[\ln\frac{g(y)}{f(y)}\Big] \leq \ln\mathbb{E}_f\Big[\frac{g(y)}{f(y)}\Big] = \ln\int g(y)\,dy = \ln 1 = 0.$$

Applied to a parametric family $f_{\boldsymbol{\theta}} = f(\cdot \mid \boldsymbol{\theta})$: if the model is correct, $\boldsymbol{\theta}_0 = \arg\min_{\boldsymbol{\theta}} \mathrm{KLIC}(f, f_{\boldsymbol{\theta}})$.

When the model is misspecified, the parametric family may still be a good **approximating model**, for example a log-normal density for wages. The natural target is then the best-fitting member of the family.

Definition 1.6 (Pseudo-true parameter)**.** $\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}\in\Theta} \mathrm{KLIC}(f, f_{\boldsymbol{\theta}})$.

Since $\mathrm{KLIC}(f, f_{\boldsymbol{\theta}}) = \int f \ln f\,dy - \mathbb{E}[\ln f(y \mid \boldsymbol{\theta})]$ and only the second term depends on $\boldsymbol{\theta}$, $\boldsymbol{\theta}^*$ maximises $\mathbb{E}[\ln f(y \mid \boldsymbol{\theta})]$. The MLE maximises the sample analogue of this expectation. ML therefore has two roles: it estimates $\boldsymbol{\theta}_0$ when the model is correct and $\boldsymbol{\theta}^*$ when it is not. In the second case we call it the **pseudo-** or **quasi-MLE** (PMLE).

Theorem 1.6 (Asymptotic distribution of the PMLE (Huber 1967; White 1982))**.** *Under regularity conditions,* $\hat{\boldsymbol{\theta}}_{PMLE} \xrightarrow{p} \boldsymbol{\theta}^*$ *and*

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_{PMLE} - \boldsymbol{\theta}^*) \xrightarrow{d} N(\mathbf{0},\ \boldsymbol{A}^{*-1}\boldsymbol{B}^*\boldsymbol{A}^{*-1}), \qquad \boldsymbol{A}^* = -\mathbb{E}[H_i(\boldsymbol{\theta}^*)], \quad \boldsymbol{B}^* = \mathbb{E}[S_i(\boldsymbol{\theta}^*)S_i(\boldsymbol{\theta}^*)'].$$

The information equality fails, so $\boldsymbol{A}^* \neq \boldsymbol{B}^*$ and only the sandwich estimator (iv) in Table 1.1 is consistent. In general, $\boldsymbol{\theta}^* \neq \boldsymbol{\theta}_0$ and the PMLE is inconsistent for the parameter we care about. There is one important exception.

1.6.1 The linear exponential family

A density belongs to the **linear exponential family** (LEF) if it can be written

$$f(y \mid \mu) = \exp\{a(\mu) + b(y) + c(\mu)\, y\}, \qquad \mathbb{E}[y] = \mu, \quad \operatorname{Var}(y) = [c'(\mu)]^{-1}.$$

Table 1.2: Leading members of the linear exponential family

Distribution	$a(\mu)$, $b(y)$, $c(\mu)$	$\mathbb{E}[y]$	$\operatorname{Var}(y)$
Normal (σ^2 known)	$-\frac{\mu^2}{2\sigma^2} - \frac{1}{2}\ln(2\pi\sigma^2)$, $-\frac{y^2}{2\sigma^2}$, $\frac{\mu}{\sigma^2}$	μ	σ^2
Bernoulli	$\ln(1-\mu)$, 0, $\ln\frac{\mu}{1-\mu}$	μ	$\mu(1-\mu)$
Exponential	$-\ln\mu$, 0, $-1/\mu$	μ	μ^2
Poisson	$-\mu$, $-\ln y!$, $\ln\mu$	μ	μ

Theorem 1.7 (Consistency of LEF-based PMLE (Gouriéroux, Monfort, and Trognon 1984))**.** *If the assumed density is LEF and the conditional mean $\mathbb{E}[y \mid \boldsymbol{x}] = m(\boldsymbol{x}, \boldsymbol{\theta}_0)$ is correctly specified, the PMLE of $\boldsymbol{\theta}$ is consistent, whatever the true distribution of y given $\boldsymbol{x}$.*

The reason is that the LEF score is proportional to $y - m(\boldsymbol{x}, \boldsymbol{\theta})$, so its expectation is zero at $\boldsymbol{\theta}_0$ as soon as the mean is right. The practical consequence is important: Poisson regression consistently estimates an exponential conditional mean for *any* nonnegative outcome, count or not, provided inference uses the sandwich variance. Computer tutorial 1 illustrates this with overdispersed count data.

1.7 Hypothesis testing: the trinity

We test $H_0 : \boldsymbol{h}(\boldsymbol{\theta}_0) = \mathbf{0}$ against $H_1 : \boldsymbol{h}(\boldsymbol{\theta}_0) \neq \mathbf{0}$, where $\boldsymbol{h} : \mathbb{R}^k \to \mathbb{R}^q$ imposes $q \leq k$ restrictions with full-rank Jacobian $\boldsymbol{R}(\boldsymbol{\theta}) = \partial \boldsymbol{h}(\boldsymbol{\theta})/\partial \boldsymbol{\theta}'$. Linear restrictions are the special case $\boldsymbol{h}(\boldsymbol{\theta}) = \boldsymbol{R}\boldsymbol{\theta} - \boldsymbol{r}$. For example, with $k = 3$, $H_0 : \theta_1 = 1,\ \theta_2 - \theta_3 = 2$ has $q = 2$ and

$$\boldsymbol{R} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad \boldsymbol{r} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Let $\hat{\boldsymbol{\theta}}_U$ be the unrestricted MLE and $\hat{\boldsymbol{\theta}}_R$ the MLE subject to $\boldsymbol{h}(\boldsymbol{\theta}) = \mathbf{0}$. A restriction such as $\beta_1 = c\beta_2$ can often be imposed by reparametrisation: $x_1\beta_1 + x_2\beta_2 = (c\,x_1 + x_2)\beta_2$, so the restricted model has a single regressor.

Theorem 1.8 (LR, Wald and LM tests)**.** *Under H_0 and correct specification, each of the following converges in distribution to χ^2_q:*

$$\begin{aligned} LR &= 2\big[\ell_n(\hat{\boldsymbol{\theta}}_U) - \ell_n(\hat{\boldsymbol{\theta}}_R)\big], \\ W &= \boldsymbol{h}(\hat{\boldsymbol{\theta}}_U)' \Big[\boldsymbol{R}(\hat{\boldsymbol{\theta}}_U)\, \widehat{\operatorname{Var}}(\hat{\boldsymbol{\theta}}_U)\, \boldsymbol{R}(\hat{\boldsymbol{\theta}}_U)'\Big]^{-1} \boldsymbol{h}(\hat{\boldsymbol{\theta}}_U), \\ LM &= S_n(\hat{\boldsymbol{\theta}}_R)'\, \widehat{\operatorname{Var}}(S_n(\hat{\boldsymbol{\theta}}_R))^{-1} S_n(\hat{\boldsymbol{\theta}}_R), \qquad \widehat{\operatorname{Var}}(S_n(\hat{\boldsymbol{\theta}}_R)) = \mathcal{I}(\hat{\boldsymbol{\theta}}_R) \text{ or } -H_n(\hat{\boldsymbol{\theta}}_R). \end{aligned}$$

The three statistics are asymptotically equivalent under H_0 and under local alternatives.

The intuition differs (Figure 1.2):

- **LR**: if the restriction is valid, imposing it should not reduce the maximised log-likelihood by much.
- **Wald**: if the restriction is valid, $\boldsymbol{h}(\hat{\boldsymbol{\theta}}_U)$ should be close to zero, because $\hat{\boldsymbol{\theta}}_U$ is consistent.
- **LM (score)**: if the restriction is valid, the slope of the log-likelihood at $\hat{\boldsymbol{\theta}}_R$ should be close to zero, because $\hat{\boldsymbol{\theta}}_R$ is then close to the unrestricted maximum.

For a single coefficient, $H_0 : \theta_j = 0$, the Wald statistic is the square of the usual t-ratio $\hat{\theta}_j / \mathrm{se}(\hat{\theta}_j)$.

i Why χ^2? Wald and LR derivations

Wald. Let $\sqrt{n}(\hat{\boldsymbol{\theta}}_U - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Omega})$ with $\boldsymbol{\Omega} = \mathcal{I}_1(\boldsymbol{\theta}_0)^{-1}$. A mean-value expansion gives $\boldsymbol{h}(\hat{\boldsymbol{\theta}}_U) = \boldsymbol{h}(\boldsymbol{\theta}_0) + \boldsymbol{R}(\boldsymbol{\theta}^*)(\hat{\boldsymbol{\theta}}_U - \boldsymbol{\theta}_0)$ with $\boldsymbol{\theta}^*$ between $\hat{\boldsymbol{\theta}}_U$ and $\boldsymbol{\theta}_0$. Under H_0, $\boldsymbol{h}(\boldsymbol{\theta}_0) = \mathbf{0}$, so by the delta method

$$\sqrt{n}\,\boldsymbol{h}(\hat{\boldsymbol{\theta}}_U) \xrightarrow{d} N\big(\mathbf{0},\ \boldsymbol{R}(\boldsymbol{\theta}_0)\boldsymbol{\Omega}\boldsymbol{R}(\boldsymbol{\theta}_0)'\big),$$

and the quadratic form $n\,\boldsymbol{h}(\hat{\boldsymbol{\theta}}_U)'\big[\boldsymbol{R}\boldsymbol{\Omega}\boldsymbol{R}'\big]^{-1}\boldsymbol{h}(\hat{\boldsymbol{\theta}}_U) \xrightarrow{d} \chi^2_q$. Replacing $\boldsymbol{R}(\boldsymbol{\theta}_0)$ by $\boldsymbol{R}(\hat{\boldsymbol{\theta}}_U)$ and $\boldsymbol{\Omega}$ by the consistent estimator $n\widehat{\mathrm{Var}}(\hat{\boldsymbol{\theta}}_U)$ does not change the limit; this gives W.
LR (scalar θ, $H_0 : \theta = \theta_0$). A second-order expansion of $\ell_n(\theta_0)$ around $\hat{\theta}$, using $S_n(\hat{\theta}) = 0$:

$$\ell_n(\theta_0) = \ell_n(\hat{\theta}) + \tfrac{1}{2}H_n(\theta^*)(\theta_0 - \hat{\theta})^2 \quad \Longrightarrow \quad LR = 2\big[\ell_n(\hat{\theta}) - \ell_n(\theta_0)\big] = \Big[-\tfrac{1}{n}H_n(\theta^*)\Big]\Big[\sqrt{n}(\hat{\theta} - \theta_0)\Big]^2.$$

The first factor converges to $\mathcal{I}_1(\theta_0)$ and $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \mathcal{I}_1^{-1})$, so $LR \xrightarrow{d} \mathcal{I}_1 \cdot \mathcal{I}_1^{-1}\chi^2_1 = \chi^2_1$. The same argument shows $LR - W \xrightarrow{p} 0$.

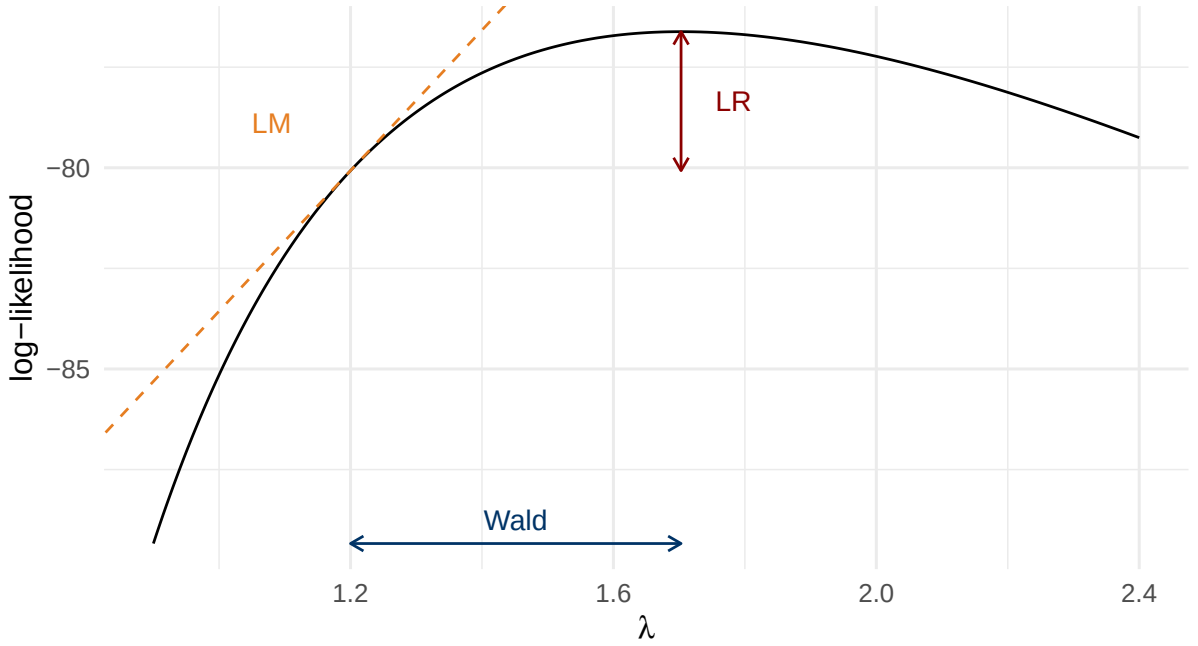


Figure 1.2: The trinity for the exponential model, testing $H_0 : \lambda = 1.2$ with n = 50 simulated observations. LR compares heights, Wald compares horizontal distance, LM compares the slope at the restricted value.

The three statistics for this sample:

```r
# y: 50 draws from an exponential with lambda = 1.5 (set.seed(42)); H0: lambda
↪  = 1.2
info <- function(l) n / l^2                  # Fisher information of the sample
LR <- 2 * (ll(l_hat) - ll(l0))
W  <- (l_hat - l0)^2 * info(l_hat)           # uses Var(l_hat) = l_hat^2 / n
LM <- slope^2 / info(l0)                     # score and information at l0
round(c(LR = LR, Wald = W, LM = LM, crit_5pct = qchisq(0.95, 1)), 3)
```

```
#>        LR      Wald        LM crit_5pct
#>     6.907     4.361     8.782     3.841
```

They differ in finite samples but lead to the same decision here. Which to use in practice?

1. **Computation.** LR needs both models, Wald only the unrestricted one, LM only the restricted one. LM is attractive when the restricted model is much simpler, e.g. when the restriction makes a nonlinear model linear.
2. **Invariance.** Only LR is invariant to how the model and the restriction are parametrised. The Wald statistic for $H_0 : \theta = 1$ and for the equivalent $H_0 : 1/\theta = 1$ can differ substantially in small samples (Cameron and Trivedi 2005, Ch. 7).
3. **Robustness.** Under misspecification, LR and the information-based LM are no longer χ^2. The Wald test with the sandwich variance remains valid, which is why robust Wald tests dominate applied work.

1.8 Computer tutorial 1

Learning goals. Write a log-likelihood as an R function and maximise it with `optim`; compute Hessian, OPG and sandwich variance estimators by hand and check them against `glm` and `sandwich`; see in a Monte Carlo study why pseudo-ML needs robust standard errors.

Packages. `sandwich`, `lmtest`, `MASS`. **Data.** `mroz.csv` (Mroz 1987), 753 married women in 1975.

1.8.1 Problem 1: Logit model for labour force participation

Variable	Description
`inlf`	= 1 if in labour force, 1975
`hours`	hours worked, 1975
`kidslt6`	number of kids under 6
`kidsge6`	number of kids aged 6–18
`age`	woman's age in years
`educ`	years of schooling
`wage`	estimated hourly wage from earnings (0 if not working)
`faminc`	family income, 1975
`exper`	actual labour market experience
`expersq`	experience squared

a) Create non-wife income in thousand dollars, `nwifeinc` = (`faminc` − `wage` × `hours`)/1000.

b) Estimate the linear probability model (LPM)

$$\texttt{inlf} = \beta_0 + \beta_1 \texttt{nwifeinc} + \beta_2 \texttt{educ} + \beta_3 \texttt{exper} + \beta_4 \texttt{expersq} + \beta_5 \texttt{age} + \beta_6 \texttt{kidslt6} + \beta_7 \texttt{kidsge6} + \varepsilon.$$

Why are heteroskedasticity-robust standard errors required here? How many fitted values lie outside $[0, 1]$?

c) Write down the logit model and its log-likelihood formally. Derive the score and the Hessian of observation i.

d) Estimate the logit model with the regressors from (b) in two ways.

1. Code the **negative** log-likelihood as an R function `negll(b, y, X)` and minimise it with `optim(..., method = "BFGS", hessian = TRUE)`, using the LPM coefficients as starting values. Supply the analytical gradient as well.
2. Use `glm(..., family = binomial(link = "logit"))`.

e) Compute the variance matrix of $\hat{\boldsymbol{\beta}}$ in four ways: (i) from the Hessian returned by `optim`; (ii) the inverse of the negative analytical Hessian; (iii) the inverse of the OPG; (iv) the sandwich. Compare with `vcov()` and `sandwich()` applied to the `glm` object.

f) Compute the average partial effects (APEs) of `educ` and `kidslt6` from the logit model and compare them with the LPM coefficients.

1.8.2 Problem 2: PML vs. ML, a Monte Carlo study

The goal is to see how ML behaves when the likelihood is misspecified but the conditional mean is right (see "Misspecification and pseudo-ML" in Chapter 1 of the lecture notes).

Correct density.

a) Simulate $n = 1000$ observations from $y_i \mid \boldsymbol{x}_i \sim \text{Poisson}(\lambda_i)$ with $\lambda_i = \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})$, $(\beta_0, \beta_1, \beta_2) = (0.2, 0.1, 0)$ and x_1, x_2 independent $N(0, 1)$.

b) Estimate $\boldsymbol{\beta}$ with `glm(..., family = poisson)`.

c) Using the score and Hessian of the Poisson model from Problem Set 1, Problem 3, compute Hessian-based and sandwich standard errors by hand. Check them against `glm` and `coeftest(..., vcov = sandwich)`.

Misspecified density.

d) Now draw y_i from a negative binomial distribution with the same mean $\mu_i = \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})$ and variance $\text{Var}(y_i \mid \boldsymbol{x}_i) = \mu_i + \mu_i^2/\theta$ with $\theta = 3$.

e) Estimate the mean function by (i) Poisson PML with Hessian-based variance, (ii) Poisson PML with sandwich variance, (iii) negative binomial ML (`MASS::glm.nb`) with Hessian-based variance.

f) Repeat (d)–(e) for 1000 Monte Carlo samples and fill in the table:

	Poisson PML (Hessian)	Poisson PML (sandwich)	NegBin ML (Hessian)
MC mean of $\hat{\beta}_2$			
MC mean of s.e.$(\hat{\beta}_2)$			

	Poisson PML (Hessian)	Poisson PML (sandwich)	NegBin ML (Hessian)
MC std. dev. of $\hat{\beta}_2$			
Rejection rate of $H_0 : \beta_2 = 0$ at 5%			

g) Discuss your results.

1.9 Exercises

Problem Set 1 contains the analytical exercises for this chapter:

- **Problem 1**: a regular one-parameter model from start to finish (score, information, bias and the Cramér–Rao bound, asymptotics, invariance) and the three tests, including the non-invariance of the Wald test.
- **Problem 2**: the shifted exponential distribution as a non-regular case, with an n-rate, non-normal limit.
- **Problem 3**: Poisson regression, its score and Hessian; with a binary regressor in closed form, with Poisson-based and robust standard errors under overdispersion. Computer tutorial 1, Problem 2 builds on it.
- **Problem 4**: canonical links in the linear exponential family.

2 Method of Moments and GMM

Readings

Cameron and Trivedi (2005), Ch. 6.1–6.5; B. E. Hansen (2022a), Ch. 12 (instrumental variables) and Ch. 13 (GMM) and 17.36–17.42 (dynamic panels); B. E. Hansen (2022b), Ch. 11 (method of moments); Wooldridge (2010), Ch. 8 and 14. Potential outcomes and weighting (Section 2.9): Imbens and Wooldridge (2009), Sections 1–5; Wooldridge (2010), Ch. 21.1–21.3. Applications: Arellano and Bond (1991) and Imai and Ratkovic (2014).

Maximum likelihood requires the full distribution of the data. Many economic models deliver less: a set of expectations that should be zero at the true parameter value. The method of moments (MM) and its generalization, the generalized method of moments (GMM), turn such moment conditions directly into estimators. OLS, IV, 2SLS and ML are all special cases, and the GMM framework gives a single recipe for their asymptotic distribution and for efficient estimation when there are more conditions than parameters (L. P. Hansen 1982).

2.1 Moment conditions

Let $\boldsymbol{w}_i$ collect all observed variables for unit i (outcomes, regressors, instruments) and let $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^k$.

Definition 2.1 (Moment function)**.** A $q \times 1$ vector function $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})$ with $q \geq k$ is a **moment function** if

$$\mathbb{E}\big[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)\big] = \mathbf{0}.$$

This equation is a (population) **moment condition** or moment restriction.

A moment function depends only on the data and the parameters, it has at least as many elements as there are parameters, and its expectation is zero at the true value. Three examples recur throughout the chapter.

Example 2.1 (Familiar estimators as moment estimators)**.**

1. **Mean.** $\psi(y_i, \mu) = y_i - \mu$. The sample analogue $\frac{1}{n}\sum_i (y_i - \hat{\mu}) = 0$ gives $\hat{\mu} = \bar{y}$.
2. **OLS.** In $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + u_i$, the assumption $\mathbb{E}[u_i \mid \boldsymbol{x}_i] = 0$ implies the k conditions $\mathbb{E}[\boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})] = \mathbf{0}$. Solving the sample analogue gives $\hat{\boldsymbol{\beta}} = \big(\sum_i \boldsymbol{x}_i\boldsymbol{x}_i'\big)^{-1}\sum_i \boldsymbol{x}_i y_i$.
3. **Maximum likelihood.** Under correct specification the score has mean zero (Theorem 1.2). With $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}) = \partial \ln f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\theta})/\partial\boldsymbol{\theta}$, the MM estimator is the root of the score, i.e. the MLE.

2.2 The method of moments estimator

Definition 2.2 (Method of moments estimator)**.** If $q = k$, the MM estimator $\hat{\boldsymbol{\theta}}$ solves the k sample moment conditions

$$\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) \equiv \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{\psi}(\boldsymbol{w}_i, \hat{\boldsymbol{\theta}}) = \mathbf{0}.$$

MM estimators are consistent whenever a law of large numbers makes the sample moments converge to the population moments and $\boldsymbol{\theta}_0$ is the unique solution of $\mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})] = \mathbf{0}$. For OLS,

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + \Big(\tfrac{1}{n}\textstyle\sum_i \boldsymbol{x}_i\boldsymbol{x}_i'\Big)^{-1}\tfrac{1}{n}\textstyle\sum_i \boldsymbol{x}_i u_i \;\xrightarrow{p}\; \boldsymbol{\beta} + \mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}\mathbb{E}[\boldsymbol{x}_i u_i] = \boldsymbol{\beta},$$

provided $\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ is nonsingular.

Theorem 2.1 (Asymptotic distribution of the MM estimator)**.** *Under regularity conditions and i.i.d. sampling,*

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0},\; \boldsymbol{A}_0^{-1}\boldsymbol{V}_0\boldsymbol{A}_0^{-1\prime}), \qquad \boldsymbol{A}_0 = \mathbb{E}\Big[\frac{\partial \boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}'}\Big], \quad \boldsymbol{V}_0 = \mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)'].$$

Sketch. A mean-value expansion of $\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \mathbf{0}$ around $\boldsymbol{\theta}_0$ gives $\mathbf{0} = \bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0) + \boldsymbol{A}_n(\bar{\boldsymbol{\theta}})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$ with $\boldsymbol{A}_n = \frac{1}{n}\sum_i \partial\boldsymbol{\psi}_i/\partial\boldsymbol{\theta}'$. Hence $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = -\boldsymbol{A}_n^{-1}\sqrt{n}\,\bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0)$. The LLN gives $\boldsymbol{A}_n \xrightarrow{p} \boldsymbol{A}_0$, the CLT gives $\sqrt{n}\,\bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{V}_0)$, and Slutsky does the rest. This is the same argument as for the MLE in Chapter 1; the MLE simply has $\boldsymbol{A}_0 = -\mathcal{I}_1$ and $\boldsymbol{V}_0 = \mathcal{I}_1$.

The variance is estimated by the sandwich $\hat{\boldsymbol{A}}^{-1}\hat{\boldsymbol{V}}\hat{\boldsymbol{A}}^{-1\prime}$ with $\hat{\boldsymbol{A}} = \frac{1}{n}\sum_i \partial\boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})/\partial\boldsymbol{\theta}'$ and $\hat{\boldsymbol{V}} = \frac{1}{n}\sum_i \boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})\boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})'$.

Example 2.2 (OLS under heteroskedasticity)**.** With $\boldsymbol{\psi} = \boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})$ we get $\boldsymbol{A}_0 = -\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ and $\boldsymbol{V}_0 = \mathbb{E}[u_i^2\boldsymbol{x}_i\boldsymbol{x}_i']$, so

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} N(\mathbf{0},\; \mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}\mathbb{E}[u_i^2\boldsymbol{x}_i\boldsymbol{x}_i']\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}).$$

White's heteroskedasticity-robust variance estimator is the MM sandwich for OLS. Likewise, the PML sandwich in Section 1.6 is the MM sandwich with the score as moment function.

2.3 The generalized method of moments

When $q > k$ the model is **overidentified**: in general no $\boldsymbol{\theta}$ sets all q sample moments to zero at once. GMM instead makes them as small as possible in a quadratic metric.

Example 2.3 (Two sets of conditions for one regression)**.** If, in addition to $\mathbb{E}[u_i \mid \boldsymbol{x}_i] = 0$, the errors are conditionally symmetric, then $\mathbb{E}[u_i^3 \mid \boldsymbol{x}_i] = 0$ and

$$\mathbb{E}\begin{pmatrix} \boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta}) \\ \boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^3 \end{pmatrix} = \mathbf{0},$$

which gives $2k$ conditions for k parameters.

Definition 2.3 (GMM estimator)**.** Let $\boldsymbol{W}_n$ be a $q \times q$ symmetric positive definite weighting matrix, possibly data dependent. The GMM estimator is

$$\hat{\boldsymbol{\theta}}(\boldsymbol{W}_n) = \arg\min_{\boldsymbol{\theta}\in\Theta} \; Q_n(\boldsymbol{\theta}), \qquad Q_n(\boldsymbol{\theta}) = \bar{\boldsymbol{g}}_n(\boldsymbol{\theta})' \, \boldsymbol{W}_n \, \bar{\boldsymbol{g}}_n(\boldsymbol{\theta}).$$

The first-order conditions are

$$\boldsymbol{A}_n(\hat{\boldsymbol{\theta}})' \, \boldsymbol{W}_n \, \bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \boldsymbol{0}, \qquad \boldsymbol{A}_n(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial \boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'},$$

i.e. GMM sets k linear combinations of the q sample moments to zero. If $q = k$, $\boldsymbol{A}_n$ is square and (generically) invertible, so the FOC reduce to $\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \boldsymbol{0}$: in the just-identified case GMM is MM, whatever $\boldsymbol{W}_n$.

2.4 Asymptotic properties of GMM

Theorem 2.2 (Consistency and asymptotic normality of GMM)**.** *Assume i.i.d. data and*

(i) ***identification:*** $\mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})] = \boldsymbol{0}$ *if and only if* $\boldsymbol{\theta} = \boldsymbol{\theta}_0$*;*

(ii) $\boldsymbol{A}_0 = \mathbb{E}[\partial \boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)/\partial \boldsymbol{\theta}']$ *is finite with full column rank* k*;*

(iii) $\boldsymbol{W}_n \xrightarrow{p} \boldsymbol{W}_0$*, symmetric positive definite;*

(iv) $\sqrt{n}\,\bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0) \xrightarrow{d} N(\boldsymbol{0}, \boldsymbol{V}_0)$*,* $\boldsymbol{V}_0 = \mathbb{E}[\boldsymbol{\psi}_i \boldsymbol{\psi}_i']$ *positive definite;*

plus standard regularity conditions (compact Θ*, uniform convergence, smoothness). Then* $\hat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}_0$ *and*

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N\Big(\boldsymbol{0}, \; (\boldsymbol{A}_0'\boldsymbol{W}_0\boldsymbol{A}_0)^{-1}\boldsymbol{A}_0'\boldsymbol{W}_0\boldsymbol{V}_0\boldsymbol{W}_0\boldsymbol{A}_0(\boldsymbol{A}_0'\boldsymbol{W}_0\boldsymbol{A}_0)^{-1}\Big).$$

The proof follows the MM sketch: expand the FOC around $\boldsymbol{\theta}_0$ and solve, $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \approx -(\boldsymbol{A}_0'\boldsymbol{W}_0\boldsymbol{A}_0)^{-1}\boldsymbol{A}_0'\boldsymbol{W}_0\sqrt{n}\,\bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0)$. Two features are worth stressing. First, **the weighting matrix does not affect consistency**, only the asymptotic variance. Second, condition (ii) is the local version of identification: if $\boldsymbol{A}_0$ has deficient rank, the moments do not pin down all parameters, which is the essence of weak or irrelevant instruments.

The variance is estimated by replacing $\boldsymbol{A}_0, \boldsymbol{V}_0, \boldsymbol{W}_0$ by $\hat{\boldsymbol{A}} = \boldsymbol{A}_n(\hat{\boldsymbol{\theta}})$, $\hat{\boldsymbol{V}} = \frac{1}{n}\sum_i \boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})\boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})'$ and $\boldsymbol{W}_n$, and dividing by n.

2.5 Efficient GMM

Theorem 2.3 (Optimal weighting matrix)**.** *Among all choices of* $\boldsymbol{W}_0$*,* $\boldsymbol{W}_0^* = \boldsymbol{V}_0^{-1}$ *minimizes the asymptotic variance (in the matrix sense of Chapter 1). The minimum is*

$$\operatorname{Avar}\left(\hat{\boldsymbol{\theta}}(\boldsymbol{V}_0^{-1})\right) = \left(\boldsymbol{A}_0'\boldsymbol{V}_0^{-1}\boldsymbol{A}_0\right)^{-1}.$$

Proof. Write $\boldsymbol{B} = (\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{V}^{-1}$ and $\boldsymbol{C} = (\boldsymbol{A}'\boldsymbol{W}\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{W} - \boldsymbol{B}$ (dropping subscripts). The asymptotic variance with weight $\boldsymbol{W}$ is $(\boldsymbol{B}+\boldsymbol{C})\boldsymbol{V}(\boldsymbol{B}+\boldsymbol{C})'$. The cross term vanishes because $\boldsymbol{C}\boldsymbol{V}\boldsymbol{B}' = [(\boldsymbol{A}'\boldsymbol{W}\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{W}\boldsymbol{A} - (\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A}](\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A})^{-1} = \boldsymbol{0}$. Hence the variance equals $(\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A})^{-1} + \boldsymbol{C}\boldsymbol{V}\boldsymbol{C}'$, and $\boldsymbol{C}\boldsymbol{V}\boldsymbol{C}'$ is positive semidefinite. □

A fact that is useful for such comparisons: for symmetric positive definite $\boldsymbol{P}$ and $\boldsymbol{Q}$, $\boldsymbol{P} - \boldsymbol{Q}$ is positive semidefinite if and only if $\boldsymbol{Q}^{-1} - \boldsymbol{P}^{-1}$ is. Comparing variances is therefore equivalent to comparing information (precision) matrices.

Intuitively, moments that are measured with more noise get less weight. "Optimal" is relative to a given set of moment conditions: poorly chosen moments remain inefficient even with the best weight. Adding valid moment conditions never increases the asymptotic variance of efficient GMM.

$\boldsymbol{V}_0$ is unknown, so efficient GMM is implemented in steps:

1. Estimate $\hat{\boldsymbol{\theta}}^{(1)}$ with a convenient weight, e.g. $\boldsymbol{W}_n = \boldsymbol{I}$ or, in linear IV, $(\boldsymbol{Z}'\boldsymbol{Z}/n)^{-1}$. It is consistent but inefficient.
2. Compute $\hat{\boldsymbol{V}} = \frac{1}{n}\sum_i \boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}}^{(1)})\boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}}^{(1)})'$ and re-estimate with $\boldsymbol{W}_n = \hat{\boldsymbol{V}}^{-1}$. This **two-step GMM** estimator is asymptotically efficient.

Repeating step 2 until convergence gives **iterated GMM**; letting the weight depend on $\boldsymbol{\theta}$ inside the objective, $Q_n(\boldsymbol{\theta}) = \bar{\boldsymbol{g}}_n(\boldsymbol{\theta})'\hat{\boldsymbol{V}}(\boldsymbol{\theta})^{-1}\bar{\boldsymbol{g}}_n(\boldsymbol{\theta})$, gives the **continuously updated estimator** (CUE). All three are asymptotically equivalent. In finite samples, estimating $\boldsymbol{V}_0$ introduces bias, especially with many moment conditions (Altonji and Segal 1996), so efficient GMM need not beat a simple one-step estimator in practice.

2.6 Testing overidentifying restrictions

GMM consistency rests on $\mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)] = \boldsymbol{0}$. With $q > k$ there is information left to check it: at the efficient estimate the sample moments should be close to zero.

Theorem 2.4 (Hansen's J test (L. P. Hansen 1982))**.** *Let $\hat{\boldsymbol{\theta}}$ be the efficient GMM estimator and $\hat{\boldsymbol{V}}$ a consistent estimator of $\boldsymbol{V}_0$. Under $H_0 : \mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)] = \boldsymbol{0}$,*

$$J = n\,\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}})'\,\hat{\boldsymbol{V}}^{-1}\,\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) \;\xrightarrow{d}\; \chi^2_{q-k}.$$

Three remarks. The factor n matters: without it the statistic converges to zero. The degrees of freedom are the number of overidentifying restrictions, $q - k$; in a just-identified model $J = 0$ by construction and nothing can be tested. Finally, the test has power only against violations that cannot be absorbed by $\boldsymbol{\theta}$: if all instruments are invalid in the same direction, J may not reject. A rejection says that *some* moment conditions fail, not which.

2.7 Conditional moments and optimal instruments

Models are often stated as a conditional restriction $\mathbb{E}[\rho(\boldsymbol{w}_i, \boldsymbol{\theta}_0) \mid \boldsymbol{z}_i] = 0$, e.g. $\rho = y_i - \boldsymbol{x}_i'\boldsymbol{\beta}$ with exogenous $\boldsymbol{z}_i$. By the law of iterated expectations, *any* function $\boldsymbol{D}(\boldsymbol{z}_i)$ yields valid unconditional moments $\mathbb{E}[\boldsymbol{D}(\boldsymbol{z}_i)\rho(\boldsymbol{w}_i, \boldsymbol{\theta}_0)] = \boldsymbol{0}$, so there are infinitely many GMM estimators. The asymptotically best choice, the **optimal instrument**, is

$$\boldsymbol{D}^*(\boldsymbol{z}_i) = \mathbb{E}\Big[\frac{\partial \rho(\boldsymbol{w}_i, \boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} \,\Big|\, \boldsymbol{z}_i\Big] \Big/ \operatorname{Var}\left(\rho(\boldsymbol{w}_i, \boldsymbol{\theta}_0) \mid \boldsymbol{z}_i\right).$$

In the linear model with exogenous regressors ($\boldsymbol{z}_i = \boldsymbol{x}_i$) this is $-\boldsymbol{x}_i/\sigma_i^2$: the efficient moment estimator is weighted least squares (GLS). With endogenous regressors it is $-\mathbb{E}[\boldsymbol{x}_i \mid \boldsymbol{z}_i]/\sigma_i^2$, which motivates 2SLS: the first stage approximates $\mathbb{E}[\boldsymbol{x}_i \mid \boldsymbol{z}_i]$ by a linear projection.

2.8 Linear instrumental variables as GMM

Consider $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + u_i$ with $\boldsymbol{x}_i$ of dimension k and some regressors correlated with u_i (omitted variables, measurement error, simultaneity). OLS is inconsistent. Let $\boldsymbol{z}_i$ be an $l \times 1$ vector of instruments, containing the exogenous regressors, with

- **exogeneity:** $\mathbb{E}[\boldsymbol{z}_i u_i] = \boldsymbol{0}$;
- **relevance:** $\operatorname{rank} \mathbb{E}[\boldsymbol{z}_i \boldsymbol{x}_i'] = k$ (requires $l \geq k$), and $\mathbb{E}[\boldsymbol{z}_i \boldsymbol{z}_i']$ nonsingular.

The moment function is $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\beta}) = \boldsymbol{z}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})$, with sample analogue $\frac{1}{n}\boldsymbol{Z}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$ in matrix notation. Minimizing $\frac{1}{n^2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$ gives the closed form

$$\hat{\boldsymbol{\beta}}(\boldsymbol{W}_n) = \left(\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X}\right)^{-1}\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{y}.$$

Here $\boldsymbol{A}_0 = -\mathbb{E}[\boldsymbol{z}_i \boldsymbol{x}_i']$ and $\boldsymbol{V}_0 = \mathbb{E}[u_i^2 \boldsymbol{z}_i \boldsymbol{z}_i']$. The familiar estimators are special cases.

Table 2.1: Linear GMM estimators

Case	Weight $\boldsymbol{W}_n$	Estimator
Just identified, $l = k$	any	IV: $\hat{\boldsymbol{\beta}}_{IV} = (\boldsymbol{Z}'\boldsymbol{X})^{-1}\boldsymbol{Z}'\boldsymbol{y}$
Overidentified	$(\boldsymbol{Z}'\boldsymbol{Z})^{-1}$	2SLS: $\left[\boldsymbol{X}'\boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}\boldsymbol{Z}'\boldsymbol{X}\right]^{-1}\boldsymbol{X}'\boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}\boldsymbol{Z}'\boldsymbol{y}$
Overidentified, homoskedastic	$(\hat{\sigma}^2\boldsymbol{Z}'\boldsymbol{Z}/n)^{-1}$ (optimal)	2SLS
Overidentified, heteroskedastic	$\left(\frac{1}{n}\sum_i \hat{u}_i^2 \boldsymbol{z}_i \boldsymbol{z}_i'\right)^{-1}$ (optimal)	two-step efficient GMM

2SLS is the GMM estimator with $\boldsymbol{W}_n = (\boldsymbol{Z}'\boldsymbol{Z})^{-1}$; under homoskedasticity, $\boldsymbol{V}_0 = \sigma^2\mathbb{E}[\boldsymbol{z}_i\boldsymbol{z}_i']$ and this weight is optimal. Under heteroskedasticity both 2SLS and two-step GMM are consistent, GMM is more efficient, and **2SLS standard errors must be heteroskedasticity robust**:

$$\widehat{\operatorname{Var}}(\hat{\boldsymbol{\beta}}) = \left(\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X}\right)^{-1}\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\Big(\sum_i \hat{u}_i^2 \boldsymbol{z}_i\boldsymbol{z}_i'\Big)\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X}\left(\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X}\right)^{-1}.$$

For two-step GMM this simplifies to $\left[\boldsymbol{X}'\boldsymbol{Z}\left(\sum_i \hat{u}_i^2 \boldsymbol{z}_i\boldsymbol{z}_i'\right)^{-1}\boldsymbol{Z}'\boldsymbol{X}\right]^{-1}$.

Weak instruments. If the instruments are only weakly correlated with the endogenous regressors, $\boldsymbol{A}_0$ is close to rank deficient. IV and 2SLS are then biased towards OLS and their normal approximation is poor, even in large samples (Staiger and Stock 1997). Always report the first-stage regression and the F statistic for the excluded instruments; a common rule of thumb flags $F < 10$ as weak.

2.8.1 Illustration: 2SLS versus efficient GMM

The simulation below has one endogenous regressor, three instruments and errors whose variance depends on the first instrument. Both estimators are consistent, but efficient GMM puts less weight on the noisy moment $\mathbb{E}[z_{1i}u_i] = 0$.

```r
set.seed(2027)
one_sample <- function(n = 500) {
  Z <- cbind(1, matrix(rnorm(3 * n), n))              # constant + 3
↪  instruments
  v <- rnorm(n)
  x <- drop(Z[, 2:4] %*% c(0.4, 0.4, 0.4)) + v
  u <- (0.5 * v + sqrt(0.75) * rnorm(n)) * exp(Z[, 2]) # endogenous,
↪  heteroskedastic
  y <- 1 + x + u
  X <- cbind(1, x)
  gmm <- function(W) solve(t(X) %*% Z %*% W %*% t(Z) %*% X, t(X) %*% Z %*% W
↪  %*% t(Z) %*% y)
  b_2sls <- gmm(solve(crossprod(Z)))                   # step 1: 2SLS
  S      <- crossprod(Z * drop(y - X %*% b_2sls)) / n
  b_gmm  <- gmm(solve(S))                              # step 2: efficient GMM
  e      <- drop(y - X %*% b_gmm)
  gbar   <- colMeans(Z * e)
  J      <- n * drop(t(gbar) %*% solve(crossprod(Z * e) / n) %*% gbar)
  c(b_2sls = b_2sls[2], b_gmm = b_gmm[2], J_reject = J > qchisq(0.95, df = 2))
}
mc <- t(replicate(1000, one_sample()))
round(c(mean_2sls = mean(mc[, 1]), mean_gmm = mean(mc[, 2]),
        sd_2sls = sd(mc[, 1]), sd_gmm = sd(mc[, 2]),
        J_rejection_rate = mean(mc[, 3])), 3)
```

```
#>        mean_2sls         mean_gmm          sd_2sls           sd_gmm
#>            1.016            1.002            0.273            0.199
#> J_rejection_rate
#>            0.039
```

Both estimators are centred on the true slope of one. The standard deviation of efficient GMM is about a quarter smaller than that of 2SLS. The J test, with $q - k = 4 - 2 = 2$ degrees of freedom, rejects the true moment conditions at roughly the nominal 5% rate.

2.9 Two-step estimators as joint GMM

Many estimators proceed in two steps: a first step estimates a nuisance parameter $\boldsymbol{\alpha}$ (a propensity score, a first-stage regression, a selection equation), and a second step estimates the parameter of interest τ given $\hat{\boldsymbol{\alpha}}$. Treating $\hat{\boldsymbol{\alpha}}$ as known in the second step gives wrong standard errors. Stacking both steps into one MM problem gives the correct ones automatically. We first introduce the leading example of this course, the average treatment effect estimated by inverse probability weighting, and then derive the general result.

2.9.1 Background: potential outcomes and the average treatment effect

Potential outcomes. Let $D_i \in \{0, 1\}$ indicate whether unit i receives a treatment (a training programme, smoking during pregnancy, a policy). Each unit has two *potential outcomes*: Y_{1i} if

treated and Y_{0i} if not (Rubin 1974). The individual effect is $Y_{1i} - Y_{0i}$. We observe only one of the two,

$$Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i},$$

so individual effects are never observed. We therefore target averages, most often the **average treatment effect**

$$\tau = \mathbb{E}[Y_{1i} - Y_{0i}] = \mu_1 - \mu_0, \qquad \mu_d = \mathbb{E}[Y_{di}],$$

or the average treatment effect on the treated, $\tau_T = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1]$. The notation implicitly assumes that a unit's outcome does not depend on the treatment of other units.

Why a simple comparison fails. The difference in observed means decomposes as

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] = \underbrace{\mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1]}_{\tau_T} + \underbrace{\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]}_{\text{selection bias}}.$$

If treated and untreated units differ in their untreated outcomes, e.g. mothers who smoke also differ in age, education and marital status, the comparison is biased. Under random assignment D_i is independent of (Y_{0i}, Y_{1i}), the selection bias is zero and $\tau_T = \tau$.

Selection on observables. Without randomization, suppose we observe covariates $\boldsymbol{x}_i$ that capture all differences between the groups that are relevant for the potential outcomes:

- *Unconfoundedness:* $(Y_{0i}, Y_{1i}) \perp D_i \mid \boldsymbol{x}_i$. Within cells of $\boldsymbol{x}_i$, treatment is as good as random.
- *Overlap:* $0 < p(\boldsymbol{x}) < 1$ for all $\boldsymbol{x}$ in the support, where $p(\boldsymbol{x}_i) = P(D_i = 1 \mid \boldsymbol{x}_i)$ is the **propensity score**.

Unconfoundedness is an identifying assumption and cannot be tested with the data; overlap can be checked by comparing the estimated propensity scores of treated and controls.

Identification by weighting. Under these two assumptions

$$\mu_1 = \mathbb{E}\Big[\frac{D_i Y_i}{p(\boldsymbol{x}_i)}\Big], \qquad \mu_0 = \mathbb{E}\Big[\frac{(1 - D_i) Y_i}{1 - p(\boldsymbol{x}_i)}\Big], \qquad \tau = \mathbb{E}\Big[\frac{D_i Y_i}{p(\boldsymbol{x}_i)} - \frac{(1 - D_i) Y_i}{1 - p(\boldsymbol{x}_i)}\Big]. \tag{2.1}$$

The proof uses $D_i Y_i = D_i Y_{1i}$, the law of iterated expectations and unconfoundedness; it is Problem Set 2, Problem 4(a). The intuition: a treated unit with propensity score 0.2 represents five units in the population, of which only one was treated. Weighting each treated unit by $1/p(\boldsymbol{x}_i)$ and each control by $1/(1 - p(\boldsymbol{x}_i))$ creates a pseudo-population in which treatment no longer depends on $\boldsymbol{x}_i$. This idea goes back to survey sampling (Horvitz and Thompson 1952); Rosenbaum and Rubin (1983) showed that the propensity score is the key balancing score.

Estimation. Replacing expectations by sample means gives the **inverse probability weighting (IPW)** estimator

$$\hat{\tau}_{IPW} = \frac{1}{n} \sum_{i=1}^{n} \Big[\frac{D_i Y_i}{\hat{p}(\boldsymbol{x}_i)} - \frac{(1 - D_i) Y_i}{1 - \hat{p}(\boldsymbol{x}_i)}\Big].$$

In finite samples the weights $D_i/\hat{p}(\boldsymbol{x}_i)$ need not average to one. The **normalized** version divides by the sum of the weights within each group,

$$\hat{\tau}_{norm} = \frac{\sum_i D_i Y_i / \hat{p}(\boldsymbol{x}_i)}{\sum_i D_i / \hat{p}(\boldsymbol{x}_i)} - \frac{\sum_i (1 - D_i) Y_i / (1 - \hat{p}(\boldsymbol{x}_i))}{\sum_i (1 - D_i) / (1 - \hat{p}(\boldsymbol{x}_i))},$$

and is usually more stable. With limited overlap some $\hat{p}(\boldsymbol{x}_i)$ are close to 0 or 1, a few observations get very large weights, and both estimators become imprecise. Regression adjustment, $\tau = \mathbb{E}[\mathbb{E}(Y_i \mid D_i = 1, \boldsymbol{x}_i) - \mathbb{E}(Y_i \mid D_i = 0, \boldsymbol{x}_i)]$, and doubly robust estimators that combine both models are the main alternatives; Imbens and Wooldridge (2009) survey them.

Why this is a two-step problem. The propensity score is unknown. In practice it is modelled parametrically, typically as a logit $p(\boldsymbol{x}_i;\boldsymbol{\alpha}) = \Lambda(\boldsymbol{x}_i'\boldsymbol{\alpha})$, and $\boldsymbol{\alpha}$ is estimated by ML in a first step. The ML first-order condition is a moment condition (Example 2.1), so both steps are MM problems:

$$\boldsymbol{\psi}_1(\boldsymbol{w}_i,\boldsymbol{\alpha}) = (D_i - \Lambda(\boldsymbol{x}_i'\boldsymbol{\alpha}))\boldsymbol{x}_i, \qquad \psi_2(\boldsymbol{w}_i,\boldsymbol{\alpha},\tau) = \frac{D_iY_i}{\Lambda(\boldsymbol{x}_i'\boldsymbol{\alpha})} - \frac{(1-D_i)Y_i}{1-\Lambda(\boldsymbol{x}_i'\boldsymbol{\alpha})} - \tau.$$

The question is how the estimation error in $\hat{\boldsymbol{\alpha}}$ affects the distribution of $\hat{\tau}$.

2.9.2 Stacking the two steps

Let $\boldsymbol{\psi}_1(\boldsymbol{w}_i,\boldsymbol{\alpha})$ (k_1 conditions) define the first step and $\psi_2(\boldsymbol{w}_i,\boldsymbol{\alpha},\tau)$ the second. The stacked moment function

$$\boldsymbol{\psi}(\boldsymbol{w}_i,\boldsymbol{\theta}) = \begin{pmatrix} \boldsymbol{\psi}_1(\boldsymbol{w}_i,\boldsymbol{\alpha}) \\ \psi_2(\boldsymbol{w}_i,\boldsymbol{\alpha},\tau) \end{pmatrix}, \qquad \boldsymbol{\theta} = (\boldsymbol{\alpha}',\tau)',$$

is just identified, and $\boldsymbol{A}_0$ is block lower triangular:

$$\boldsymbol{A}_0 = \begin{pmatrix} \boldsymbol{A}_{11} & \boldsymbol{0} \\ \boldsymbol{A}_{21} & A_{22} \end{pmatrix}, \quad \boldsymbol{A}_{11} = \mathbb{E}\Big[\frac{\partial \boldsymbol{\psi}_1}{\partial \boldsymbol{\alpha}'}\Big],\ \boldsymbol{A}_{21} = \mathbb{E}\Big[\frac{\partial \psi_2}{\partial \boldsymbol{\alpha}'}\Big],\ A_{22} = \mathbb{E}\Big[\frac{\partial \psi_2}{\partial \tau}\Big].$$

Applying Theorem 2.1 and reading off the last row gives the influence representation

$$\sqrt{n}(\hat{\tau} - \tau_0) = -\frac{1}{A_{22}}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\Big[\psi_{2i} - \boldsymbol{A}_{21}\boldsymbol{A}_{11}^{-1}\boldsymbol{\psi}_{1i}\Big] + o_p(1).$$

The term $\boldsymbol{A}_{21}\boldsymbol{A}_{11}^{-1}\boldsymbol{\psi}_{1i}$ is the **correction for the estimated first step**. It vanishes only if $\boldsymbol{A}_{21} = \boldsymbol{0}$, i.e. if the second-step moment does not depend on $\boldsymbol{\alpha}$ in expectation. Solving the stacked system jointly gives the same point estimates as the sequential procedure, because the first block does not involve τ; only the standard errors change.

For IPW, $A_{22} = -1$ and $\boldsymbol{A}_{21} \neq \boldsymbol{0}$. A surprising but general feature: when the first step is efficient ML of a correctly specified propensity score model, the correction *reduces* the variance, so using the estimated propensity score is more efficient than using the true one (Hirano, Imbens, and Ridder 2003). The naive standard error that treats $\hat{p}(\boldsymbol{x}_i)$ as known is therefore conservative for IPW, but in other two-step problems it can be too small.

```
set.seed(2027)
one_draw <- function(n = 1000) {
  x  <- rnorm(n)
  p  <- plogis(-0.3 + 0.8 * x)            # true propensity score
  d  <- rbinom(n, 1, p)
  y  <- 1 + 2 * x + d * 1 + rnorm(n)       # true ATE = 1
  ph <- fitted(glm(d ~ x, family = binomial))
  ipw <- function(p) mean(d * y / p - (1 - d) * y / (1 - p))
  c(true_p = ipw(p), estimated_p = ipw(ph))
}
sim <- replicate(2000, one_draw())
round(rbind(mean = rowMeans(sim), sd = apply(sim, 1, sd)), 3)
```

```
#>      true_p estimated_p
#> mean  1.000       1.002
#> sd    0.183       0.086
```

Both versions are centred on the true ATE of one, and the estimator with the *estimated* propensity score has the clearly smaller standard deviation.

Problem Set 2, Problem 4 derives the moment functions and the blocks of $\boldsymbol{A}_0$ for the average treatment effect on the treated, and Computer Tutorial 2 estimates the model jointly, computes the stacked sandwich variance and compares it with the bootstrap.

2.10 Further applications

GMM is the natural framework whenever theory delivers orthogonality conditions rather than a full likelihood. Two examples from the slides:

- **Dynamic panel data.** In $y_{it} = \gamma y_{i,t-1} + \boldsymbol{x}_{it}'\boldsymbol{\beta} + \eta_i + \varepsilon_{it}$, first differencing removes η_i but makes $\Delta y_{i,t-1}$ correlated with $\Delta\varepsilon_{it}$. Lags $y_{i,t-2}, y_{i,t-3}, \ldots$ are valid instruments, and their number grows with t. Arellano and Bond (1991) use all of them in an efficient GMM estimator.
- **Covariate balancing propensity score.** Imai and Ratkovic (2014) estimate the propensity score by GMM, combining the logit score with moment conditions that require the weighted covariate means to balance between treated and controls. The overidentified version allows a J test of the propensity score model.

2.11 Computer tutorial 2

Learning goals. Estimate a two-step estimator as one joint MM problem with `nleqslv`; compute its sandwich variance by hand and compare with the bootstrap; implement 2SLS, efficient two-step GMM and the J test in matrix form and check them against `AER::ivreg`.

Packages. `nleqslv`, `boot`, `sandwich`, `lmtest`, `AER`. **Data.** `cattaneo2.csv` (Cattaneo 2010), 4642 births; `card1995.csv` (Card 1995), 3613 young men from the NLS 1976 cross-section.

2.11.1 Problem 1: The effect of smoking on birth weight (IPW as joint MM)

This problem uses the IPW results of Section 2.9 of the lecture notes and Problem Set 2, Problem 4(a).

Variable	Description
`bweight`	infant birth weight (grams)
`mbsmoke`	= 1 if the mother smoked during pregnancy
`mmarried`	= 1 if the mother is married
`mage`	mother's age (years)
`medu`	mother's education (years)
`fbaby`	= 1 if first baby

We want the average treatment effect τ of `mbsmoke` on `bweight`. The propensity score is modelled as a logit,

$$P(\texttt{mbsmoke} = 1 \mid X_i) = \Lambda\big(\alpha_0 + \alpha_1 \texttt{mmarried} + \alpha_2 \texttt{mage} + \alpha_3 \texttt{mage}^2 + \alpha_4 \texttt{fbaby} + \alpha_5 \texttt{medu}\big).$$

a) Compute the raw difference in mean birth weight between smokers and non-smokers. Why is it unlikely to be the ATE?

b) Write the stacked moment function for $\boldsymbol{\theta} = (\boldsymbol{\alpha}', \tau)'$ from Section 2.9.2 of the lecture notes as an R function that returns the $k+1$ sample moments, and solve the system with `nleqslv`. Compare $\hat{\boldsymbol{\alpha}}$ with `glm` and $\hat{\tau}$ with the IPW formula evaluated at the `glm` propensity score. Why do they coincide?

c) Estimate the asymptotic variance of $\hat{\boldsymbol{\theta}}$ with the MM sandwich $\hat{\boldsymbol{A}}^{-1}\hat{\boldsymbol{V}}\hat{\boldsymbol{A}}^{-1\prime}/n$, computing $\hat{\boldsymbol{A}}$ analytically. Report the standard error of $\hat{\tau}$ and compare it with the "naive" standard error that treats the propensity score as known.

d) Compute the ATE with normalized weights (weights that sum to one within each treatment group).

e) Estimate the standard errors of both ATE estimators by the nonparametric bootstrap (500 replications). Remember that each bootstrap replication has to re-estimate the propensity score.

2.11.2 Problem 2: Returns to schooling with college proximity (IV and GMM)

Variable	Description
`lwage76`	log hourly wage, 1976
`ed76`	years of education, 1976
`age76`	age in 1976
`black`, `reg76r`, `smsa76r`	= 1 if black; lives in the South; lives in an urban area (1976)
`nearc4a`, `nearc4b`	= 1 if grew up near a 4-year public / private college

Following Card (1995) and B. E. Hansen (2022a, Ch. 12), consider

$$\log w_i = \beta_0 + \beta_1\,\texttt{educ}_i + \beta_2\,\texttt{exper}_i + \beta_3\,\texttt{exper}_i^2/100 + \beta_4\texttt{black}_i + \beta_5\texttt{south}_i + \beta_6\texttt{urban}_i + u_i,$$

with potential experience `exper` = `age` − `educ` − 6. Education is endogenous (ability is omitted), and so is experience, because it is a function of education. The excluded instruments are `nearc4a`, `nearc4b`, `age76` and `age76^2/100`.

a) Construct `exper` and `exper2`, drop observations with missing values and estimate the equation by OLS.

b) Estimate the first stage for `ed76`. Test the joint significance of `nearc4a` and `nearc4b` with a heteroskedasticity-robust F test. Are the instruments strong?

c) Count the moment conditions q and parameters k. Compute 2SLS with the GMM formula and $\boldsymbol{W}_n = (\boldsymbol{Z}'\boldsymbol{Z})^{-1}$, with heteroskedasticity-robust standard errors. Check your results with `AER::ivreg` and `sandwich`.

d) Compute the two-step efficient GMM estimator and its standard errors.

e) Compute Hansen's J statistic and its p-value. What does it test here?

f) Compare the estimated returns to schooling across OLS, 2SLS and GMM.

2.12 Exercises

Problem Set 2 contains the analytical exercises for this chapter:

- **Problem 1**: two MM estimators of an exponential mean, their asymptotic distributions, efficient GMM combining both moments, and why the identity weighting matrix depends on the units of measurement.
- **Problem 2**: ML, MM and GMM for the beta distribution.
- **Problem 3**: one endogenous regressor and two instruments, by hand: IV, GMM as a weighted average of the IV estimators, 2SLS, efficient GMM, robust standard errors and the J test.
- **Problem 4**: identification of the ATE and the ATT by inverse probability weighting, the ATT as a joint MM problem, and covariate balancing. Computer tutorial 2, Problem 1 implements the ATE version.

Part II

Computation and flexible models

3 Simulation-Based Estimation

> **Readings**
>
> Cameron and Trivedi (2005), Ch. 12 (simulation-based methods) and Ch. 11 (bootstrap); B. E. Hansen (2022a), Ch. 10 (resampling); Train (2009), Ch. 9 and 10.

Many likelihood functions and moment conditions contain integrals without a closed form: random effects in nonlinear panel models, random coefficients, unobserved heterogeneity, multinomial probit choice probabilities. This chapter shows how to approximate such integrals, deterministically by quadrature or stochastically by simulation, and what happens to the statistical properties of an estimator when the exact integral is replaced by an approximation. The last section uses the same simulation idea for inference: the bootstrap.

3.1 Integrals in econometric models

Two examples recur.

Random effects. In a panel with individual effect $c_i \sim g(c \mid \boldsymbol{\eta})$, the likelihood contribution of individual i is

$$f(\boldsymbol{y}_i \mid \boldsymbol{X}_i; \boldsymbol{\beta}, \boldsymbol{\eta}) = \int f(\boldsymbol{y}_i \mid \boldsymbol{X}_i, \boldsymbol{\beta}, c)\, g(c \mid \boldsymbol{\eta})\, dc,$$

a one-dimensional integral for every i and every trial value of the parameters.

Discrete choice. In a multinomial probit with m alternatives the choice probability is an $(m-1)$-dimensional integral of a multivariate normal density over a rectangle.

In both cases the object we need is an **expectation**,

$$I = \mathbb{E}[h(X)] = \int h(x)\, g(x)\, dx, \qquad X \sim g, \tag{3.1}$$

where h and g are known for given parameter values. A plain integral $\int_a^b f(x)\, dx$ is the special case $h = (b-a)f$ and g the uniform density on $[a, b]$.

3.2 Deterministic integration: quadrature

Quadrature approximates an integral by a weighted sum of function values at chosen **nodes** x_j with **weights** ω_j,

$$\hat{I} = \sum_{j=1}^{m} \omega_j\, f(x_j).$$

Newton–Cotes rules use equally spaced nodes on $[a, b]$. The midpoint rule evaluates f at the midpoints of n intervals with weights $(b-a)/n$; the trapezoidal rule averages the function values at the two ends of each interval; Simpson's rule fits a quadratic through three neighbouring

points. Their error shrinks at a polynomial rate in n (e.g. n^{-2} for the midpoint rule if f'' is bounded).

Gaussian quadrature chooses nodes and weights optimally for a given weight function $w(x)$:

$$\int_c^d w(x)\, r(x)\, dx \approx \sum_{j=1}^{m} \omega_j\, r(x_j).$$

With m nodes the rule is **exact if r is a polynomial of degree $2m-1$ or less**. Nodes and weights are tabulated and available in software (`statmod::gauss.quad` in R). The common cases are Gauss–Legendre ($w=1$ on $[-1,1]$), Gauss–Laguerre ($w=e^{-x}$ on $(0,\infty)$) and **Gauss–Hermite** ($w=e^{-x^2}$ on $\mathbb{R}$).

Example 3.1 (Gauss–Hermite for a normal expectation)**.** Let $X \sim N(\mu,\sigma^2)$. The substitution $x=\mu+\sqrt{2}\sigma z$ gives

$$\mathbb{E}[h(X)] = \int h(x)\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/(2\sigma^2)}dx = \frac{1}{\sqrt{\pi}}\int e^{-z^2}h(\mu+\sqrt{2}\sigma z)\,dz \approx \frac{1}{\sqrt{\pi}}\sum_{j=1}^{m}\omega_j\, h(\mu+\sqrt{2}\sigma z_j).$$

In the random effects probit with $c_i \sim N(0,\sigma_c^2)$ this gives the likelihood contribution $\ell_i \approx \pi^{-1/2}\sum_j \omega_j f(\boldsymbol{y}_i \mid \boldsymbol{X}_i, \boldsymbol{\beta}, \sqrt{2}\sigma_c z_j)$, which is how `xtprobit` in Stata computes it.

Table 3.1: Gauss<U+2013>Hermite nodes and weights for $w(z)=e^{-z^2}$ (only $z_j \geq 0$; the rule is symmetric).

m	node	weight
2	0.707107	0.886227
3	0.000000	1.181636
3	1.224745	0.295409
4	0.524648	0.804914
4	1.650680	0.081313
5	0.000000	0.945309
5	0.958572	0.393619
5	2.020183	0.019953

Choosing m. There is a trade-off between speed and accuracy. In practice one re-estimates with more nodes and checks that the results do not change (Stata's `quadchk`).

Many dimensions. A d-dimensional integral can be handled by a product rule, applying a one-dimensional rule in each coordinate. This needs m^d function evaluations: with $m=10$ nodes and $d=5$ dimensions already 10^5 per observation. This **curse of dimensionality** is the main reason to simulate.

3.3 Monte Carlo integration

The integral in Equation 3.1 is a population mean. Draw $x_1,\dots,x_S$ independently from g and use the sample mean:

$$\hat{I}_S = \frac{1}{S}\sum_{s=1}^{S} h(x_s).$$

Theorem 3.1 (Direct Monte Carlo integration)**.** *If* $\mathbb{E}|h(X)| < \infty$*, then* $\hat{I}_S$ *is unbiased,* $\mathbb{E}[\hat{I}_S] = I$*, and* $\hat{I}_S \xrightarrow{p} I$ *as* $S \to \infty$*. If moreover* $\sigma_h^2 = \mathrm{Var}(h(X)) < \infty$*,*

$$\mathrm{Var}(\hat{I}_S) = \frac{\sigma_h^2}{S}, \qquad \sqrt{S}(\hat{I}_S - I) \xrightarrow{d} N(0, \sigma_h^2).$$

These are the WLLN and the CLT applied to the draws. Three consequences:

- The **simulation standard error** $\hat{\sigma}_h/\sqrt{S}$, with $\hat{\sigma}_h$ the standard deviation of the $h(x_s)$, is computed from the same draws. Report it.
- The error shrinks at rate $S^{-1/2}$ **whatever the dimension of** X. This is why simulation wins in high dimensions, and loses to quadrature in one or two smooth dimensions.
- Halving the error requires four times as many draws.

Draws from g are usually generated from uniforms by the **inverse transform**: if $U \sim U(0,1)$ and G is the cdf of X, then $G^{-1}(U) \sim G$.

Example 3.2 (A normal expectation with a known answer)**.** Let $X \sim N(0,1)$ and $I = \mathbb{E}[\Phi(0.5 + X)]$. Since $\Phi(0.5 + X) = P(Z \le 0.5 + X \mid X)$ for an independent $Z \sim N(0,1)$, $I = P(Z - X \le 0.5) = \Phi(0.5/\sqrt{2}) = 0.638163$. Figure 3.1 compares Monte Carlo, Halton draws (Section 3.3.2) and Gauss–Hermite quadrature. Computer Tutorial 3 asks you to reproduce it.

3.3.1 Variance reduction

The variance σ_h^2/S can be reduced without more draws by using draws that are not independent or not from g.

Importance sampling. Rewrite the integral with a proposal density p that is positive wherever $hg \neq 0$:

$$I = \int \frac{h(x)g(x)}{p(x)}\, p(x)\, dx = \mathbb{E}_p[w(X)h(X)], \qquad w(x) = \frac{g(x)}{p(x)}, \qquad \hat{I}_{IS} = \frac{1}{S}\sum_{s=1}^{S} w(x_s)h(x_s), \quad x_s \sim p.$$

$\hat{I}_{IS}$ is unbiased for any valid p, but its variance depends on p. A good proposal puts its mass where $|h(x)|g(x)$ is large and avoids huge weights. If $h \ge 0$, the choice $p \propto hg$ would give zero variance; it is infeasible, because its normalizing constant is I, but it shows the direction. Importance sampling is the natural tool for small probabilities and for densities that are hard to draw from.

Example 3.3 (A tail probability)**.** For $Y \sim N(0,1)$, $p_0 = P(Y \ge 3) = 0.00135$. With $S = 100$ direct draws, the estimate is zero in $(1 - p_0)^{100} \approx 0.874$ of all samples. With the proposal $N(\mu, 1)$ the weight is $w(y) = \phi(y)/\phi(y - \mu) = e^{-\mu y + \mu^2/2}$ and

$$\mathbb{E}_p\big[w(Y)^2 \mathbf{1}\{Y \ge 3\}\big] = e^{\mu^2}\big(1 - \Phi(3 + \mu)\big),$$

so the simulation standard deviation with $S = 100$ falls from 0.00367 ($\mu = 0$) to 3.09×10^{-4} ($\mu = 4$) and 2.48×10^{-4} ($\mu = 3$). Problem Set 3 derives this.

Antithetic draws. If g is symmetric around zero, pair each draw x_s with $-x_s$ (for uniforms: u_s with $1 - u_s$) and average $\frac{1}{2}[h(x_s) + h(-x_s)]$. With S pairs, i.e. $2S$ function evaluations,

$$\mathrm{Var}(\hat{I}_{anti}) = \frac{1}{2S}\Big[\sigma_h^2 + \mathrm{Cov}\,\big(h(X), h(-X)\big)\Big],$$

compared with $\sigma_h^2/(2S)$ for $2S$ independent draws. Antithetic draws help if the covariance is negative, which is the case for monotone h.

3.3.2 Quasi-random draws

Pseudo-random draws cluster by chance and leave gaps. **Quasi-random** (low-discrepancy) sequences are deterministic and fill $[0,1]$ evenly; they are transformed to the target distribution by the inverse cdf. The leading example is the **Halton sequence**. For a prime base b, write $i = \sum_k a_k b^k$ in base b and reflect the digits about the decimal point: $\varphi_b(i) = \sum_k a_k b^{-k-1}$. For $b = 3$ the sequence starts

$$\tfrac{1}{3},\ \tfrac{2}{3},\ \tfrac{1}{9},\ \tfrac{4}{9},\ \tfrac{7}{9},\ \tfrac{2}{9},\ \tfrac{5}{9},\ \tfrac{8}{9},\ \ldots$$

Each new block of points falls into the gaps left by the previous ones.

```
# Halton sequence in base b (equivalent to randtoolbox::halton for dim = 1, b =
↪  2)
halton_seq <- function(n, base = 2, skip = 0) {
  sapply(seq_len(n) + skip, function(i) {
    f <- 1; r <- 0
    while (i > 0) { f <- f / base; r <- r + f * (i %% base); i <- i %/% base }
    r
  })
}
round(halton_seq(8, base = 3), 3)
```

```
#> [1] 0.333 0.667 0.111 0.444 0.778 0.222 0.556 0.889
```

For smooth integrands in low dimensions the error of quasi-Monte Carlo is close to $1/S$ instead of $1/\sqrt{S}$. In d dimensions one uses a different prime for each coordinate; with many dimensions the coordinates of the early points become correlated, and scrambled or randomized versions are preferred. In estimation, one long sequence is generated and consecutive blocks are used for different observations; the first points are often discarded.

The figure shows the three rates: the Monte Carlo error falls by about one decimal for a hundred-fold increase in S, Halton draws roughly ten times faster, and Gauss–Hermite reaches machine precision with about a dozen nodes, because the integrand is smooth and one-dimensional.

3.4 Maximum simulated likelihood

Suppose the density of $\boldsymbol{y}_i$ given $\boldsymbol{x}_i$ is an integral over unobservables $\boldsymbol{u}$ with known density g,

$$f(\boldsymbol{y}_i \mid \boldsymbol{x}_i; \boldsymbol{\theta}) = \int \tilde{f}(\boldsymbol{y}_i \mid \boldsymbol{x}_i, \boldsymbol{u}; \boldsymbol{\theta})\, g(\boldsymbol{u})\, d\boldsymbol{u}.$$

Draw $\boldsymbol{u}_{i1}, \ldots, \boldsymbol{u}_{iS}$ from g for each i and replace f by the unbiased simulator $\hat{f}_i(\boldsymbol{\theta}) = S^{-1} \sum_s \tilde{f}(\boldsymbol{y}_i \mid \boldsymbol{x}_i, \boldsymbol{u}_{is}; \boldsymbol{\theta})$.

Definition 3.1 (Maximum simulated likelihood estimator)**.**

$$\hat{\boldsymbol{\theta}}_{MSL} = \arg\max_{\boldsymbol{\theta}} \ \frac{1}{n} \sum_{i=1}^{n} \ln \hat{f}_i(\boldsymbol{\theta}), \qquad \hat{f}_i(\boldsymbol{\theta}) = \frac{1}{S} \sum_{s=1}^{S} \tilde{f}(\boldsymbol{y}_i \mid \boldsymbol{x}_i, \boldsymbol{u}_{is}; \boldsymbol{\theta}).$$

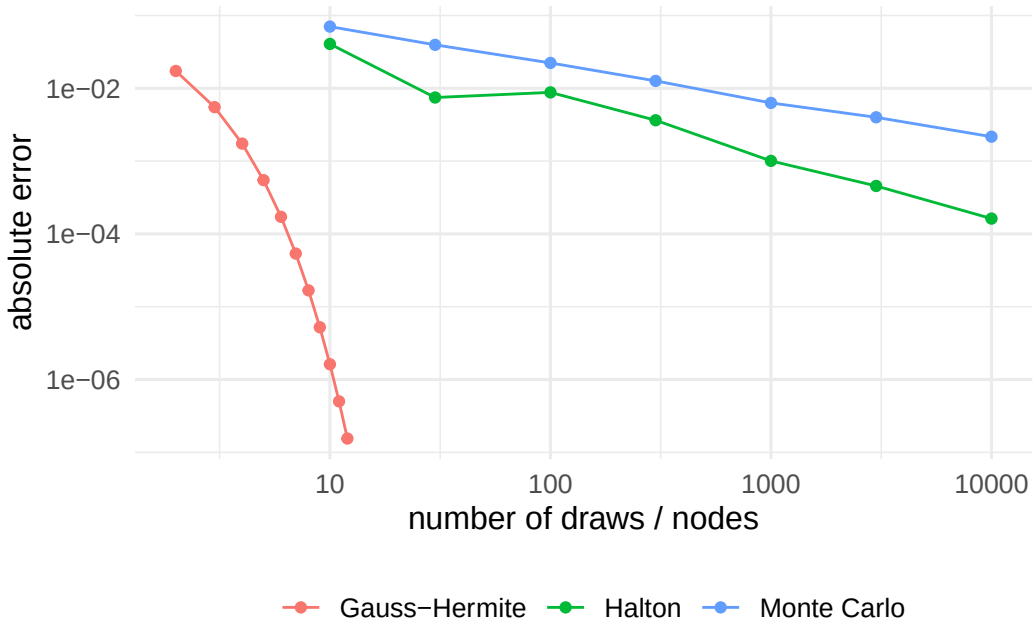


Figure 3.1: Absolute error in approximating $\mathbb{E}[\Phi(0.5+X)]$, $X \sim N(0,1)$: Monte Carlo (average over 200 replications), Halton draws and Gauss–Hermite quadrature. Log scales.

If the distribution of $\boldsymbol{u}$ depends on parameters (e.g. $c_i \sim N(0, \sigma_c^2)$), write $\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{\varepsilon}; \boldsymbol{\theta})$ with draws $\boldsymbol{\varepsilon}$ from a fixed distribution ($c_{is} = \sigma_c \varepsilon_{is}$ with $\varepsilon_{is} \sim N(0,1)$).

Keep the draws fixed. The draws are generated once, before the optimization, and reused for every evaluation of the objective and its derivatives. New draws at every iteration would make the simulated likelihood jump between iterations and the optimizer fail to converge.

Simulation bias. $\hat{f}_i$ is unbiased for f_i, but the log is concave, so by Jensen's inequality $\mathbb{E}[\ln \hat{f}_i] < \ln f_i$. A second-order expansion gives

$$\mathbb{E}\big[\ln \hat{f}_i(\boldsymbol{\theta})\big] \approx \ln f_i(\boldsymbol{\theta}) - \frac{1}{2S} \frac{\operatorname{Var}\left(\tilde{f}(\boldsymbol{y}_i \mid \boldsymbol{x}_i, \boldsymbol{u}; \boldsymbol{\theta})\right)}{f_i(\boldsymbol{\theta})^2}.$$

The bias term depends on $\boldsymbol{\theta}$, so it shifts the maximizer: **with fixed S, MSL is inconsistent.**

Theorem 3.2 (Asymptotics of MSL (Cameron and Trivedi 2005, Ch. 12))**.** *Under the regularity conditions for ML and with independent draws across observations:*

1. *If $S \to \infty$ as $n \to \infty$, $\hat{\boldsymbol{\theta}}_{MSL}$ is consistent.*
2. *If $\sqrt{n}/S \to 0$, $\hat{\boldsymbol{\theta}}_{MSL}$ is asymptotically equivalent to the (infeasible) MLE: $\sqrt{n}(\hat{\boldsymbol{\theta}}_{MSL} - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}(\boldsymbol{\theta}_0)^{-1})$.*

In practice S is chosen by experimentation: increase S (and use Halton draws) until the estimates stop changing. Standard errors are computed as for ML, from the Hessian or the sandwich of the simulated log-likelihood; with S large enough the simulation adds nothing to first order.

3.5 Method of simulated moments

If the moment function contains an integral, $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}) = \int \tilde{\boldsymbol{\psi}}(\boldsymbol{w}_i, \boldsymbol{u}; \boldsymbol{\theta}) g(\boldsymbol{u})\, d\boldsymbol{u}$, replace it by the simulator $\hat{\boldsymbol{\psi}}_i(\boldsymbol{\theta}) = S^{-1} \sum_s \tilde{\boldsymbol{\psi}}(\boldsymbol{w}_i, \boldsymbol{u}_{is}; \boldsymbol{\theta})$ and apply (G)MM to $\frac{1}{n}\sum_i \hat{\boldsymbol{\psi}}_i(\boldsymbol{\theta})$ (McFadden 1989; Pakes and Pollard 1989).

The key difference to MSL: the simulator enters the objective **linearly**. Since $\mathbb{E}[\hat{\boldsymbol{\psi}}_i(\boldsymbol{\theta}_0)] = \mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)] = \mathbf{0}$, the simulated moment condition still holds at $\boldsymbol{\theta}_0$.

Theorem 3.3 (Properties of MSM)**.** *If the GMM estimator based on $\boldsymbol{\psi}$ is consistent and asymptotically normal and the draws are independent across i and of the data, then for **fixed** S the MSM estimator is consistent and asymptotically normal. Its asymptotic variance is that of GMM plus a term due to simulation noise, which vanishes as $S \to \infty$. With S independent draws per observation, simulation adds $\mathbb{E}\big[\operatorname{Var}(\tilde{\boldsymbol{\psi}} \mid \boldsymbol{w}_i)\big]/S$ to the variance $\boldsymbol{V}_0$ of the moment function.*

Table 3.2: MSL and MSM compared

	MSL	MSM
Consistent with fixed S	no	yes
Efficient as $S \to \infty$	yes, if $\sqrt{n}/S \to 0$	only if the moments are the efficient ones
Cost of small S	bias	variance

Example 3.4 (Unobserved heterogeneity (Cameron and Trivedi 2005, Ch. 12))**.** Let $y_i = \theta + u_i + \varepsilon_i$ with $\varepsilon_i \sim N(0,1)$ and u_i standard Gumbel (extreme value type I), $g(u) = e^{-u}\exp(-e^{-u})$, independent. The density of y_i,

$$f(y_i;\theta) = \int \phi(y_i - \theta - u)\, g(u)\, du,$$

has no closed form. With draws u_{is} from g:

- **MSL** maximizes $\frac{1}{n}\sum_i \ln\big[\frac{1}{S}\sum_s \phi(y_i - \theta - u_{is})\big]$.
- **MSM** uses $\mathbb{E}[y_i - \theta - u_i] = 0$ with $\mathbb{E}[u_i]$ replaced by the simulated mean, $\hat{\theta}_{MSM} = \bar{y} - \frac{1}{nS}\sum_i \sum_s u_{is}$.

(Here $\mathbb{E}[u_i]$ is in fact known, Euler's constant $\gamma \approx 0.5772$, so MM is feasible; the example is chosen so that we can compare.)

```
rgumbel <- function(n) -log(-log(runif(n)))
msl <- function(y, u) {                    # u: n x S matrix of draws
  obj <- function(th) mean(log(rowMeans(dnorm(y - th - u))))
  optimize(obj, c(-5, 5), maximum = TRUE)$maximum
}
msm <- function(y, u) mean(y) - mean(u)

set.seed(2027)
one_sample <- function(S, n = 100, theta = 1) {
  y <- theta + rgumbel(n) + rnorm(n)
  u <- matrix(rgumbel(n * S), n, S)        # draws fixed before optimizing
  c(msl = msl(y, u), msm = msm(y, u))
```

```
}
tab <- t(sapply(c(1, 2, 5, 20, 100, 1000), function(S) {
  r <- replicate(1000, one_sample(S))
  c(S = S, bias_MSL = mean(r["msl", ]) - 1, sd_MSL = sd(r["msl", ]),
    bias_MSM = mean(r["msm", ]) - 1, sd_MSM = sd(r["msm", ]))
}))
knitr::kable(round(tab, 3))
```

Table 3.3: MSL and MSM in the unobserved-heterogeneity model: bias and standard deviation over 1000 samples ($n = 100$, $\theta_0 = 1$).

S	bias_MSL	sd_MSL	bias_MSM	sd_MSM
1	-0.005	0.205	-0.005	0.205
2	0.145	0.191	-0.006	0.180
5	0.134	0.183	-0.006	0.170
20	0.065	0.165	-0.004	0.162
100	0.016	0.166	-0.007	0.173
1000	0.003	0.154	-0.001	0.162

The table illustrates Theorem 3.2 and Theorem 3.3:

- MSM is unbiased for every S; its standard deviation falls towards the MM value $\sqrt{(1+\pi^2/6)/n} = 0.163$. With independent draws, $\mathrm{Var}(\hat{\theta}_{MSM}) = [1 + \pi^2/6 + \pi^2/(6S)]/n$ (Problem Set 3, Problem 4, derives the analogous result for an exponential shock).
- MSL has a sizeable bias for small S that disappears only as S grows. With many draws it is slightly more precise than MSM, because it uses the whole likelihood.
- With $S = 1$ MSL and MSM coincide here: $\ln \phi$ is quadratic, so the first-order condition is linear in θ. This is special to the example.

3.6 The bootstrap

Simulation is also useful for inference. The asymptotic variance of many estimators is tedious to derive (two-step estimators, sample quantiles, functions of several estimates). The **nonparametric bootstrap** (Efron 1979) approximates the sampling distribution of an estimator $\hat{\theta} = T(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n)$ by resampling the data:

1. Draw a bootstrap sample $\boldsymbol{w}_1^*, \ldots, \boldsymbol{w}_n^*$ of size n **with replacement** from the observed data (in a regression: resample whole observations $(y_i, \boldsymbol{x}_i)$, the *pairs* bootstrap).
2. Compute $\hat{\theta}_b^* = T(\boldsymbol{w}_1^*, \ldots, \boldsymbol{w}_n^*)$, using exactly the same procedure as for $\hat{\theta}$, including all estimation steps.
3. Repeat for $b = 1, \ldots, B$.

The **bootstrap standard error** is the standard deviation of $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$. The **percentile confidence interval** uses the $\alpha/2$ and $1-\alpha/2$ quantiles of the $\hat{\theta}_b^*$. B of a few hundred suffices for standard errors; percentile intervals need more ($B \geq 999$).

The logic: the empirical distribution of the data is to the sample what the population is to the data. If $\hat{\theta}$ is asymptotically normal and smooth in the empirical distribution, the bootstrap distribution of $\sqrt{n}(\hat{\theta}^* - \hat{\theta})$ approximates that of $\sqrt{n}(\hat{\theta} - \theta_0)$ (Cameron and Trivedi 2005, Ch. 11).

Example 3.5 (Standard error of a sample median)**.** For a sample of $n = 200$ from a lognormal distribution, the asymptotic variance of the median m is $1/(4f(m)^2 n)$, which needs an estimate of the density at the median. The bootstrap avoids this:

```
set.seed(2027)
y  <- exp(rnorm(200))
bt <- replicate(999, median(sample(y, replace = TRUE)))
c(median = median(y), se_boot = sd(bt),
  se_asymptotic = 1 / (2 * dlnorm(1) * sqrt(200)))   # uses the true density
```

```
#>        median       se_boot se_asymptotic
#>    0.99025928    0.09515879    0.08862269
```

```
quantile(bt, c(0.025, 0.975))                        # percentile 95% CI
```

```
#>     2.5%    97.5%
#> 0.853481 1.197444
```

The bootstrap standard error is close to the asymptotic one, which here uses the true density at the true median and would not be available in practice.

When it fails. The bootstrap is not a cure-all. It fails or needs modification for non-smooth or boundary problems (the maximum of a sample, parameters on the boundary of the parameter space), with weak instruments, and with dependent data, where whole blocks or clusters have to be resampled. In Computer Tutorial 2 we already used it for the two-step IPW estimator: there the bootstrap must re-estimate the propensity score in every replication.

3.7 Computer tutorial 3

Learning goals. Approximate an integral by Monte Carlo, quasi-random draws and Gaussian quadrature and compare their accuracy; implement maximum simulated likelihood and the method of simulated moments and see the simulation bias of MSL; compute bootstrap standard errors and confidence intervals and check their coverage.

Packages. `statmod` (Gaussian quadrature); optionally `randtoolbox` (Halton sequences). No data set: all data are simulated.

3.7.1 Problem 1: Three ways to compute an integral

Let $X \sim N(0,1)$ and $I = \mathbb{E}[\Phi(0.5 + X)]$, where Φ is the standard normal cdf.

a) Show that $I = \Phi(0.5/\sqrt{2})$. *Hint:* write $\Phi(0.5 + X) = P(Z \le 0.5 + X \mid X)$ for $Z \sim N(0,1)$ independent of X.

b) Write a function `mc_int(S)` that approximates I by direct Monte Carlo integration with S draws and returns the estimate together with its simulation standard error. Evaluate it for $S = 10, 100, 1000, 10000$.

c) Approximate I with S Halton draws (base 2), transformed to $N(0,1)$ by the inverse cdf. Use the function `halton_seq()` from Section 3.3.2 of the lecture notes or `randtoolbox::halton()`. Compare the errors with (b).

d) Approximate I by Gauss–Hermite quadrature with $m = 2, 3, 5, 10, 20$ nodes (`statmod::gauss.quad(m, kind = "hermite")`). Mind the change of variables in Example 3.1 of the lecture notes.

e) Use antithetic pairs $(x_s, -x_s)$ with $S/2$ pairs. Is the variance smaller than with S independent draws? Why?

3.7.2 Problem 2: MSL and MSM with unobserved heterogeneity

Consider the model of Example 3.4 of the lecture notes: $y_i = \theta + u_i + \varepsilon_i$ with $\varepsilon_i \sim N(0,1)$ and u_i standard Gumbel, independent, $\theta_0 = 1$.

a) Generate a sample of $n = 100$. Draw Gumbel variables by the inverse transform $u = -\ln(-\ln U)$, $U \sim U(0,1)$.

b) Write the simulated log-likelihood as a function of θ and of an $n \times S$ matrix of draws. Compute $\hat{\theta}_{MSL}$ for $S = 1, 10, 100, 1000$. Why must the draws be generated before the optimization?

c) Compute $\hat{\theta}_{MSM}$ for the same S, and the MM estimator $\bar{y} - \gamma$, which uses the known mean $\gamma = 0.5772$ of the Gumbel distribution.

d) Repeat (b) with Halton draws: take one long Halton sequence, discard the first 10 points, and use consecutive blocks of S points for the observations.

e) Monte Carlo: for $S = 2$ and $S = 50$, repeat (a)–(c) 300 times and report the mean and standard deviation of $\hat{\theta}_{MSL}$ and $\hat{\theta}_{MSM}$. Relate the results to Theorems 3.2 and 3.3 of the lecture notes.

3.7.3 Problem 3: Bootstrap standard errors and confidence intervals

a) Draw a sample of $n = 200$ from a lognormal distribution, $y_i = e^{z_i}$ with $z_i \sim N(0,1)$. The population median is 1.

b) Compute the bootstrap standard error of the sample median with $B = 999$ and the 95% percentile confidence interval.

c) Check the coverage of the percentile interval: repeat (a)–(b) 300 times with $B = 299$ and count how often the interval contains 1.

d) Do the same for the sample **maximum** as an estimator of the upper end of the support of a $U(0,1)$ distribution. What goes wrong?

3.8 Exercises

Problem Set 3 contains the analytical exercises for this chapter:

- **Problem 1**: Monte Carlo integration of $\int_0^1 xe^x\,dx$, the number of draws needed for a given precision, and antithetic draws.
- **Problem 2**: importance sampling for a normal tail probability.
- **Problem 3**: Gauss–Hermite quadrature and exactness for polynomials.
- **Problem 4**: MSL and MSM in the unobserved-heterogeneity model of Example 3.4, with an exponential instead of a Gumbel shock, for which the exact likelihood is available as a benchmark.

4 Nonparametric Methods

In preparation

This chapter will be added before the corresponding lectures. Planned content:

- Histogram and kernel density estimation; bias, variance and bandwidth choice
- Nadaraya-Watson and local linear regression
- Rule-of-thumb and cross-validated bandwidths; confidence bands
- Curse of dimensionality and semiparametric alternatives
- Computer tutorial 4

5 Introduction to Statistical Learning

In preparation

This chapter will be added before the corresponding lectures. Planned content:

- Prediction vs. estimation; bias-variance trade-off
- Cross-validation
- Ridge, lasso and elastic net
- Trees and ensembles (overview)
- Lasso for inference and double machine learning (outlook)
- Computer tutorial 5

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A Large-Sample Asymptotics

This appendix collects the asymptotic tools used throughout the notes. They are assumed known from Research Metrics 1; the summary is for reference. Wooldridge (2010), Ch. 3, B. E. Hansen (2022a), Ch. 6 and B. E. Hansen (2022b), Ch. 7–8 give proofs.

A.1 Modes of convergence

Let $\boldsymbol{z}_n$ be a sequence of random vectors.

Definition A.1 (Convergence in probability)**.** $\boldsymbol{z}_n \xrightarrow{p} \boldsymbol{c}$ if for every $\varepsilon > 0$, $P(\|\boldsymbol{z}_n - \boldsymbol{c}\| > \varepsilon) \to 0$ as $n \to \infty$. We also write $\operatorname{plim} \boldsymbol{z}_n = \boldsymbol{c}$. An estimator $\hat{\boldsymbol{\theta}}_n$ is **consistent** if $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$.

Definition A.2 (Convergence in distribution)**.** $\boldsymbol{z}_n \xrightarrow{d} \boldsymbol{z}$ if $P(\boldsymbol{z}_n \le \boldsymbol{t}) \to P(\boldsymbol{z} \le \boldsymbol{t})$ at every continuity point $\boldsymbol{t}$ of the cdf of $\boldsymbol{z}$. The distribution of $\boldsymbol{z}$ is the **limiting distribution**.

Convergence in probability implies convergence in distribution; the converse holds only if the limit is a constant.

Order notation. $\boldsymbol{z}_n = o_p(1)$ means $\boldsymbol{z}_n \xrightarrow{p} \mathbf{0}$; $\boldsymbol{z}_n = O_p(1)$ means $\boldsymbol{z}_n$ is bounded in probability (for every ε there is an M with $P(\|\boldsymbol{z}_n\| > M) < \varepsilon$ for all n). Any $\boldsymbol{z}_n$ that converges in distribution is $O_p(1)$, and $o_p(1) \cdot O_p(1) = o_p(1)$.

A.2 Laws of large numbers and central limit theorems

Theorem A.1 (Weak law of large numbers)**.** *If $\boldsymbol{y}_1, \dots, \boldsymbol{y}_n$ are i.i.d. with $\mathbb{E}\|\boldsymbol{y}_i\| < \infty$, then $\bar{\boldsymbol{y}}_n = \frac{1}{n}\sum_i \boldsymbol{y}_i \xrightarrow{p} \mathbb{E}[\boldsymbol{y}_i]$.*

Theorem A.2 (Central limit theorem (Lindeberg–Lévy))**.** *If $\boldsymbol{y}_1, \dots, \boldsymbol{y}_n$ are i.i.d. with mean $\boldsymbol{\mu}$ and finite variance matrix $\boldsymbol{\Sigma}$, then*

$$\sqrt{n}\,(\bar{\boldsymbol{y}}_n - \boldsymbol{\mu}) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (\boldsymbol{y}_i - \boldsymbol{\mu}) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Sigma}).$$

Both apply to functions of the data: in the notes $\boldsymbol{y}_i$ is replaced by a score $S_i(\boldsymbol{\theta}_0)$, a moment function $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)$ or a simulated quantity $h(x_s)$.

A.3 Continuous mapping, Slutsky and the delta method

Theorem A.3 (Continuous mapping theorem)**.** *Let g be continuous.*

(i) If $\boldsymbol{z}_n \xrightarrow{p} \boldsymbol{c}$, then $g(\boldsymbol{z}_n) \xrightarrow{p} g(\boldsymbol{c})$.

(ii) If $\boldsymbol{z}_n \xrightarrow{d} \boldsymbol{z}$, then $g(\boldsymbol{z}_n) \xrightarrow{d} g(\boldsymbol{z})$.

For (i), continuity at $\boldsymbol{c}$ suffices.

Theorem A.4 (Slutsky's theorem)**.** *If $z_n \xrightarrow{d} z$ and $c_n \xrightarrow{p} c$ (a constant), then $z_n + c_n \xrightarrow{d} z + c$, $c_n z_n \xrightarrow{d} cz$, and $z_n / c_n \xrightarrow{d} z/c$ if $c \neq 0$.*

In matrix form: if $\boldsymbol{H}_n \xrightarrow{p} \boldsymbol{H}$ and $\boldsymbol{b}_n \xrightarrow{d} N(\boldsymbol{\mu}, \boldsymbol{B})$, then $\boldsymbol{H}_n \boldsymbol{b}_n \xrightarrow{d} N(\boldsymbol{H}\boldsymbol{\mu}, \boldsymbol{H}\boldsymbol{B}\boldsymbol{H}')$.

The matrix form is the workhorse of the notes. Every asymptotic normality proof writes $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = \boldsymbol{H}_n \boldsymbol{b}_n$, where $\boldsymbol{H}_n$ is an (inverse) average of derivatives that converges in probability by the WLLN, and $\boldsymbol{b}_n$ is $\sqrt{n}$ times an average of mean-zero terms that is asymptotically normal by the CLT.

Theorem A.5 (Delta method)**.** *If $\sqrt{n}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{V})$ and $\boldsymbol{g} : \mathbb{R}^k \to \mathbb{R}^q$ is continuously differentiable at $\boldsymbol{\beta}$ with Jacobian $\boldsymbol{G} = \partial \boldsymbol{g}(\boldsymbol{\beta}) / \partial \boldsymbol{\beta}'$, then*

$$\sqrt{n}(\boldsymbol{g}(\hat{\boldsymbol{\beta}}_n) - \boldsymbol{g}(\boldsymbol{\beta})) \xrightarrow{d} N(\mathbf{0},\ \boldsymbol{G}\boldsymbol{V}\boldsymbol{G}').$$

Why: by the mean value theorem, $\boldsymbol{g}(\hat{\boldsymbol{\beta}}_n) = \boldsymbol{g}(\boldsymbol{\beta}) + \boldsymbol{G}(\boldsymbol{\beta}^*)(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta})$ with $\boldsymbol{\beta}^*$ between $\hat{\boldsymbol{\beta}}_n$ and $\boldsymbol{\beta}$. Since $\boldsymbol{\beta}^* \xrightarrow{p} \boldsymbol{\beta}$, $\boldsymbol{G}(\boldsymbol{\beta}^*) \xrightarrow{p} \boldsymbol{G}$, and Slutsky gives the result.

Example A.1 (Delta method)**.** Let y_i be i.i.d. with mean μ and variance σ^2, and $g(\mu) = e^{\mu}$. Then $\sqrt{n}(e^{\bar{y}} - e^{\mu}) \xrightarrow{d} N(0, e^{2\mu}\sigma^2)$, and a standard error of $e^{\bar{y}}$ is $e^{\bar{y}}\hat{\sigma}/\sqrt{n}$.

A.4 Mean value expansions

Most proofs expand a first-order condition around the true value. For a differentiable vector function $\bar{\boldsymbol{g}}_n(\boldsymbol{\theta})$,

$$\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0) + \frac{\partial \bar{\boldsymbol{g}}_n(\boldsymbol{\theta}^*)}{\partial \boldsymbol{\theta}'} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0),$$

where $\boldsymbol{\theta}^* = \boldsymbol{\theta}_0 + \lambda(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$, $0 \leq \lambda \leq 1$, lies on the segment between $\hat{\boldsymbol{\theta}}$ and $\boldsymbol{\theta}_0$ (strictly, a different $\boldsymbol{\theta}^*$ for each row). If $\hat{\boldsymbol{\theta}}$ is consistent, so is $\boldsymbol{\theta}^*$. If $\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \mathbf{0}$ by definition of the estimator, solving for $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0$ gives the representation used for the MLE (Theorem 1.5), MM (Theorem 2.1) and GMM (Theorem 2.2).

A.5 Consistency of extremum estimators

ML, GMM and MSL estimators maximize or minimize a sample objective $Q_n(\boldsymbol{\theta})$. The standard argument has three ingredients:

1. **Identification:** the limit objective $Q_0(\boldsymbol{\theta}) = \operatorname{plim} Q_n(\boldsymbol{\theta})$ has a unique maximum at $\boldsymbol{\theta}_0$ (for ML this follows from the KLIC inequality; for GMM from $\mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})] = \mathbf{0}$ only at $\boldsymbol{\theta}_0$).
2. **Uniform convergence:** $\sup_{\boldsymbol{\theta} \in \Theta} |Q_n(\boldsymbol{\theta}) - Q_0(\boldsymbol{\theta})| \xrightarrow{p} 0$, which holds under compactness of Θ, continuity and a dominance condition (a uniform law of large numbers).
3. **Continuity** of Q_0.

Then $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$ (Wooldridge 2010, Thm. 12.2; B. E. Hansen 2022a, Ch. 22.4). Pointwise convergence of Q_n is not enough: the maximizer of Q_n could drift away from $\boldsymbol{\theta}_0$.

B A Short R Primer

> **In preparation**
>
> This chapter will be added before the corresponding lectures. Planned content:
>
> - Projects, scripts and working directories
> - Data import, data frames and matrices
> - Matrix algebra and OLS by hand
> - Writing functions; optimisation with optim
> - Computer tutorial 0